MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: **Mansourah Massarah Gold Project**
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: **CALCULATION**
3660 LOW GRADE ACID WASH/ELUTION/CARBON REGEN.
DESIGN OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Regen.	:	Regeneration

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1. INTENT

The purpose of this document is to analysis and design of foundations in **LOW GRADE ACID WASH/ELUTION/CARBON REGEN** area. In this document design & analysis of Tank Foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The Tank Foundation design has been carried out using excel sheets. Design data of tanks are taken from document no. MD-216-3660-EG-ME-GAD-1004-A01, MD-216-3660-EG-ME-GAD-1034-A01, MD-216-3660-EG-ME-GAD-1035-A01, MD-216-3660-EG-ME-GAD-1037-A01, MD-216-3660-EG-ME-GAD-1038-A01. Tanks for which foundation design covered in this document are 3660-TNK-001, 3660-TNK-002, 3660-TNK-004, 3660-TNK-005 & 3660-TNK-010.

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.2 Loading

A. Primary Load

LOAD 1 = TANK DEAD LOAD (D)

LOAD 2 = OPERATIONAL FLUID LOAD (F1)

LOAD 3 = TEST/MAX FLUID LOAD (F2)

LOAD 4 = WIND LOAD (W)

LOAD 5 = SEISMIC LOAD (E)

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4.3 Load combination

A. Allowable Stress Design.

101 D + F1

102 0.6D + W

103 D + F1 + W

104 D + F1 + 0.7E

105 0.6 (D + F1) + 0.7E

106 D + F2

B) Ultimate Strength Design.

1011.4 (D + F1)

102 0.9D + 1.6W

103 1.2(D + F1) +1.6 W

104 1.2(D + F1) + E

105 0.9 (D + F1) + E

106 1.4(D + F2)

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.5m depth below EGL (RL 912.200) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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4.7 Vertical Modulus of Subgrade Reaction

The Vertical Modulus of Subgrade Reaction has been considered as 40500 kN/m^2/m at 1.5m depth below EGL (RL 912.500) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

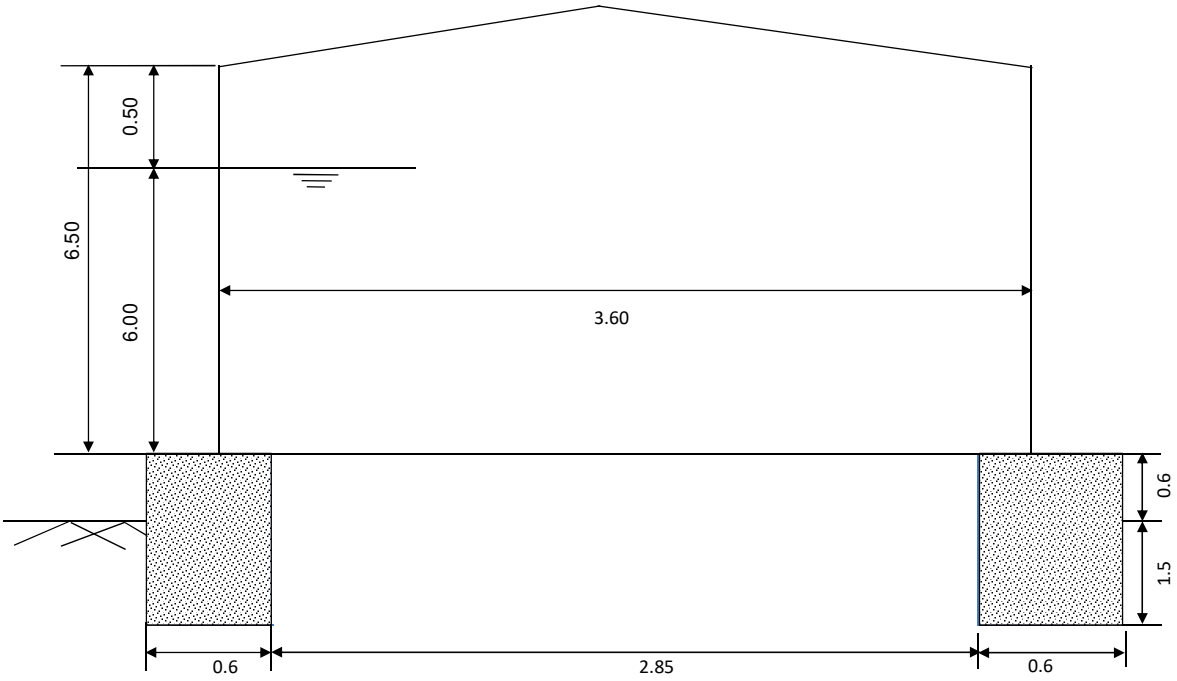
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5.1 DESIGN OF FOUNDATION FOR 3660-TNK-002/010

	Loading Type		
Weight, V	Empty (W)	70	kN
	Operating (Wo)	760	kN
	Test (Wt)	760	kN

Summary of Tank Specification



Diameter of Tank, D	=	3.6 m
Height of Tank, H	=	6.5 m
Height of Tank Fluid, H_L	=	6 m
Inner Diameter of Concrete Ring Wall, ID	=	2.85 m
Outer Diameter of Concrete Ring Wall, OD	=	4.05 m
Height of Concrete Ring Wall, h	=	2.10 m
Thickness of Concrete Ring Wall, t	=	0.6 m
Top of Concrete above Finished Ground Level	=	0.6 m
Unit Weight of Liquid, γ_L	=	11.0 kN/m^3
Unit Weight of Soil, γ_s	=	20.0 kN/m^3
Unit Weight of Concrete, γ_c	=	24.0 kN/m^3

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	24.34	79.09
	Operating	24.34	79.09
Seismic Loads	Operating	59.40	193.05

Foundation Weight

a) Ring Wall Weight = 327.76 kN

a) Inside Soil Weight = 267.93 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 6.50 m^2

Full cross-sectional area for operating condition, Ao= 12.88 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 4.92 m^3

Sectional modulus for operating condition,Zo = 6.52 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickness of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, (W/πD) = 6.19 kN/m

So, Required Min Wall THK.= 0.2516 m = 251.6 mm < 600 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301,2007 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	760.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	42.00	24.34	79.09
	Operating	LC103	D + F1 + W	760.00	24.34	79.09
		LC104	D + F1+ 0.7 E	760.00	41.58	135.14
		LC105	0.6 (D + F1) + 0.7 E	456.00	41.58	135.14
	Test	LC106	D + F2	760.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1355.69	0.00	105.24	105.24	4.28	OK
LC102	369.76	130.20	83.31	30.41	7.18	OK
LC103	1355.69	130.20	125.20	85.27	4.78	OK
LC104	1355.69	222.45	139.34	71.13	4.30	OK
LC105	1051.69	222.45	115.75	47.53	5.17	OK
LC106	1355.69	0.00	105.24	105.24	4.28	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
	V (kN)					
LC101	1355.69	0.4	542.28	0.00	-	-
LC102	369.76	0.4	147.90	24.34	6.1	OK
LC103	1355.69	0.4	542.28	24.34	22.3	OK
LC104	1355.69	0.4	542.28	41.58	13.0	OK
LC105	1051.69	0.4	420.68	41.58	10.1	OK
LC106	1355.69	0.4	542.28	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
	V (kN)					
LC101	1355.69	2.025	2745.27	0.00	-	-
LC102	369.76	2.025	748.76	130.20	5.8	OK
LC103	1355.69	2.025	2745.27	130.20	21.1	OK
LC104	1355.69	2.025	2745.27	222.45	12.3	OK
LC105	1051.69	2.025	2129.67	222.45	9.6	OK
LC106	1355.69	2.025	2745.27	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	1064.00	0.00	10.18	4.58
Short term	Empty	LC102	0.9 D + 1.6 W	63.00	126.55	10.18	4.58
	Operating	LC103	1.2 (D + F1) + 1.6 W	912.00	126.55	10.18	4.58
		LC104	1.2 (D + F1) + E	912.00	193.05	10.18	4.58
		LC105	0.9 (D + F1) + E	684.00	193.05	10.18	4.58
	Test	LC106	1.4 (D + F2)	1064.00	0.00	10.18	4.58

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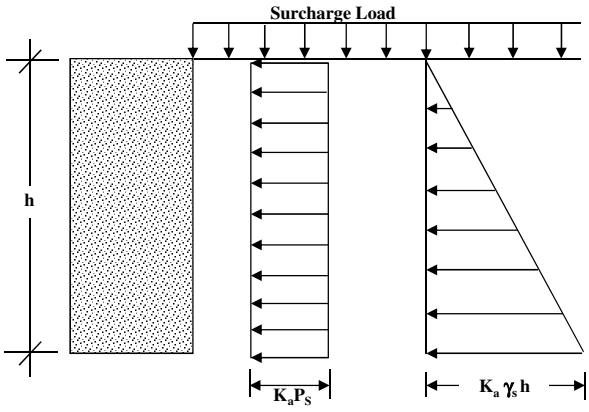
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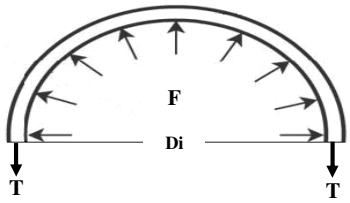
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Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	1064.00	0.00	104.53
LC102	63.00	126.55	33.82
LC103	912.00	126.55	117.23
LC104	912.00	193.05	131.74
LC105	684.00	193.05	109.35
LC106	1064.00	0.00	104.53

Maximum Surcharge Load, Ps = 131.74 kN/m^2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 65.87 KN/m^2

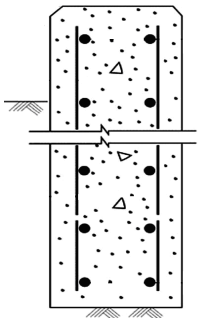
P2 = $1.4 * \gamma_S * h * K_a$ = 29.40 KN/m^2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 169.202 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 241.113 KN



Hoop Tension Bar

1. Horizontal Reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 3150 mm^2

Minimum As = 0.0025 x b x D = 3150 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 18 Nos. D16= 3266.251 mm^2 OK

2. Vertical Reinforcement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 5 each D12 (@ 200mm)

=1131mm2 > Required As OK

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Wind & Seismic Load Calculation for 3660-TNK-002 & 010

Refer MD-216-3600-EG-ME-GAD-1004 & 1035 for design data

WIND LOAD CALCULATION

Basic Wind Speed (Vs) :	44m/s	=158Km/h
Velocity pressure coeff (Kd) :	0.95	
Exposure Category :	C	
Topography factor (Kzt) :	1	
Exposure coefficient (Kz) :	1	(Refer table 7.2-2 of SBC-301)
Tank Height (H) :	6.5	m
Tank Diameter (D) :	3.6	m

Velocity presure (qz) = $q_z = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (kN/m^2); V in km/h (Ref eq. 7.2-12 of SBC-301) = 1.30kN/m2

Force coefficient, Cf = 0.8 (Refer table 7.2-15 of SBC-301)

Thus, Base shear due to wind = 1.3x0.8x6.5x3.6= 24.34 kN

Moment at base due to wind = 24.336x6.5x0.5= 79.09 kN.m

SEISMIC LOAD CALCULATION

Spectral response accelaration, S1 =	0.10g
Spectral response accelaration, Ss =	0.02g
Site Class :	B
Site coefficient, Fa =	1
Site coefficient, Fv =	1
Design Spectral response accelaration, Sd1 =	0.013g
Design Spectral response accelaration, Sds =	0.067g
Time Period (TL) :	4 sec
Liquid height, H1 =	3.9 m

The seismic base shear is the combination of the impulsive and convective components:

V = Vi + Vc

$$V_i = \frac{S_{ai} W_i}{(R/I)} \qquad V_c = \frac{S_{ac} I}{1.5} W_c$$

Conservatively considering Sai = Sds, and R = 1, I = 1.15

Hence, Vi = 58.56 kN

Natural time period for first convective mode of sloshing :

$$T_c = 2\pi \sqrt{\frac{D}{3.68\, g\, tan\, h\left(\frac{3.68H}{D}\right)}} \quad = \quad 7 \text{ sec}$$

For $T_c > T_L$:

$$S_{ac} = \frac{1.5\, S_{D1} T_L}{T_c^2} \quad = \quad 0.00159$$

Thus, Vc = 0.84 kN

Hence, seismic base shear, V = 59.4 kN

Moment at base due to seismic = 59.4x6.5x0.5= 193.05 kN.m

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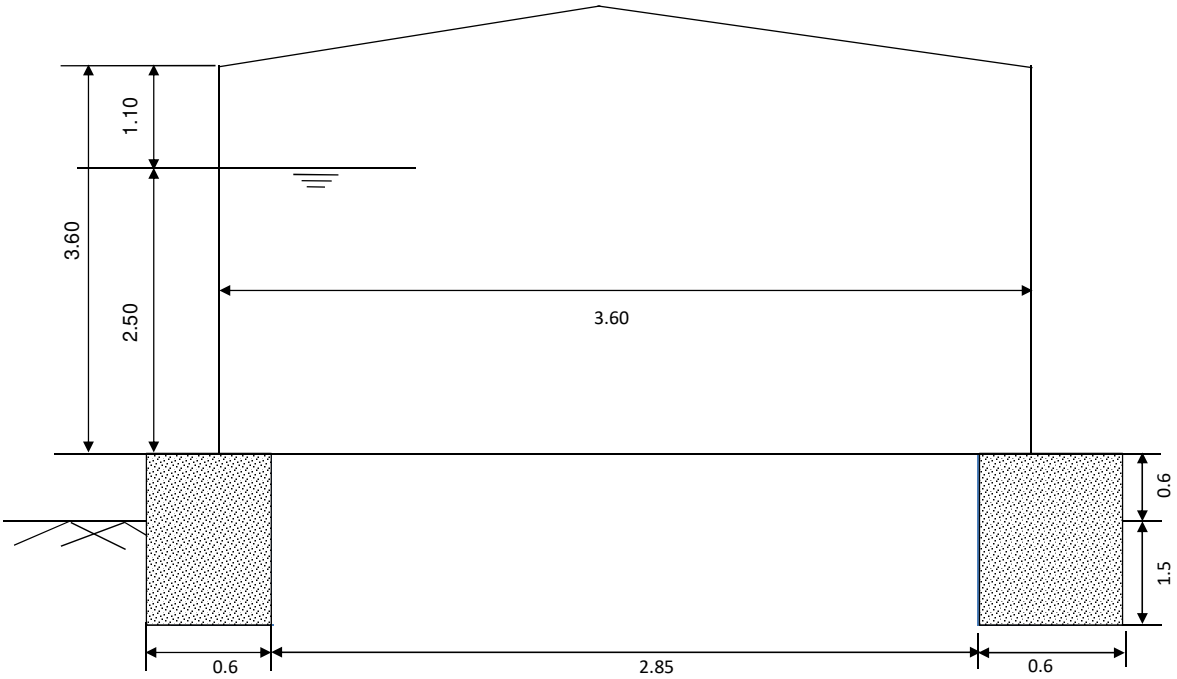
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5.2 DESIGN OF FOUNDATION FOR 3660-TNK-001/004

	Loading Type		
Weight, V	Empty (W)	38.4	kN
	Operating (Wo)	370	kN
	Test (Wt)	370	kN

Summary of Tank Specification



Diameter of Tank, D	=	3.6 m
Height of Tank, H	=	3.6 m
Height of Tank Fluid, H_L	=	2.5 m
Inner Diameter of Concrete Ring Wall, ID	=	2.85 m
Outer Diameter of Concrete Ring Wall, OD	=	4.05 m
Height of Concrete Ring Wall, h	=	2.10 m
Thickness of Concrete Ring Wall, t	=	0.6 m
Top of Concrete above Finished Ground Level	=	0.6 m
Unit Weight of Liquid, γ_L	=	11.0 kN/m^3
Unit Weight of Soil, γ_s	=	20.0 kN/m^3
Unit Weight of Concrete, γ_c	=	24.0 kN/m^3

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	12.86	23.14
	Operating	12.86	23.14
Seismic Loads	Operating	30.12	54.22

Foundation Weight

a) Ring Wall Weight = 327.76 kN

a) Inside Soil Weight = 267.93 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 6.50 m^2

Full cross-sectional area for operating condition, Ao= 12.88 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 4.92 m^3

Sectional modulus for operating condition,Zo = 6.52 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickness of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, (W/πD) = 3.40 kN/m

So, Required Min Wall THK.= 0.6346 m = 634.6 mm > 600 mm NOT OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301,2007 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	370.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	23.04	12.86	23.14
	Operating	LC103	D + F1 + W	370.00	12.86	23.14
		LC104	D + F1+ 0.7 E	370.00	21.08	37.95
		LC105	0.6 (D + F1) + 0.7 E	222.00	21.08	37.95
	Test	LC106	D + F2	370.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	965.69	0.00	74.96	74.96	6.00	OK
LC102	350.80	50.14	64.13	43.76	9.33	OK
LC103	965.69	50.14	82.65	67.27	7.24	OK
LC104	965.69	82.23	87.57	62.35	6.83	OK
LC105	817.69	82.23	76.08	50.86	7.87	OK
LC106	965.69	0.00	74.96	74.96	6.00	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
	V (kN)					
LC101	965.69	0.4	386.28	0.00	-	-
LC102	350.80	0.4	140.32	12.86	10.9	OK
LC103	965.69	0.4	386.28	12.86	30.0	OK
LC104	965.69	0.4	386.28	21.08	18.3	OK
LC105	817.69	0.4	327.08	21.08	15.5	OK
LC106	965.69	0.4	386.28	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
	V (kN)					
LC101	965.69	2.025	1955.52	0.00	-	-
LC102	350.80	2.025	710.36	50.14	14.2	OK
LC103	965.69	2.025	1955.52	50.14	39.0	OK
LC104	965.69	2.025	1955.52	82.23	23.8	OK
LC105	817.69	2.025	1655.82	82.23	20.1	OK
LC106	965.69	2.025	1955.52	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	518.00	0.00	10.18	4.58
Short term	Empty	LC102	0.9 D + 1.6 W	34.56	37.03	10.18	4.58
	Operating	LC103	1.2 (D + F1) + 1.6 W	444.00	37.03	10.18	4.58
		LC104	1.2 (D + F1) + E	444.00	54.22	10.18	4.58
		LC105	0.9 (D + F1) + E	333.00	54.22	10.18	4.58
	Test	LC106	1.4 (D + F2)	518.00	0.00	10.18	4.58

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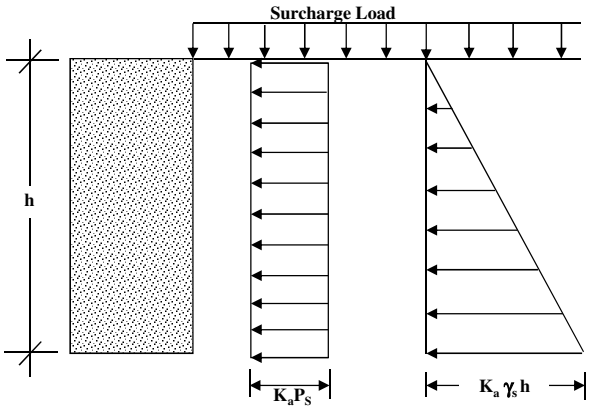
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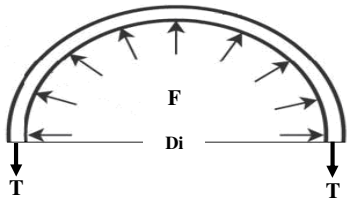
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Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	518.00	0.00	50.89
LC102	34.56	37.03	11.48
LC103	444.00	37.03	51.70
LC104	444.00	54.22	55.46
LC105	333.00	54.22	44.55
LC106	518.00	0.00	50.89

Maximum Surcharge Load, Ps = 55.46 kN/m^2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 27.73 KN/m^2

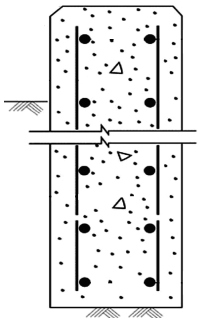
P2 = $1.4 * \gamma_S * h * K_a$ = 29.40 KN/m^2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 89.0995 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 126.967 KN



Hoop Tension Bar

1. Horizontal Reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 3150 mm^2

Minimum As = 0.0025 x b x D = 3150 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 15 Nos. D16= 2721.876 mm^2 NOT OK

2. Vertical Reinforcement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 5 each D12 (@ 200mm)

=1131mm2 > Required As OK

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Wind & Seismic Load Calculation for 3660-TNK-001 & 004

Refer MD-216-3600-EG-ME-GAD-1004 & 1035 for design data

WIND LOAD CALCULATION

Basic Wind Speed (Vs) :	43m/s	=155Km/h
Velocity pressure coeff (Kd) :	0.95	
Exposure Category :	C	
Topography factor (Kzt) :	1	
Exposure coefficient (Kz) :	1	(Refer table 7.2-2 of SBC-301)
Tank Height (H) :	3.6	m
Tank Diameter (D) :	3.6	m

Velocity presure (qz) = $q_z = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (kN/m^2); V in km/h (Ref eq. 7.2-12 of SBC-301) = 1.24kN/m2

Force coefficient, Cf = 0.8 (Refer table 7.2-15 of SBC-301)

Thus, Base shear due to wind = 1.24x0.8x3.6x3.6= 12.86 kN

Moment at base due to wind = 12.85632x3.6x0.5= 23.14 kN.m

SEISMIC LOAD CALCULATION

Spectral response accelaration, S1 =	0.10g
Spectral response accelaration, Ss =	0.02g
Site Class :	B
Site coefficient, Fa =	1
Site coefficient, Fv =	1
Design Spectral response accelaration, Sd1 =	0.013g
Design Spectral response accelaration, Sds =	0.067g
Time Period (TL) :	4 sec
Liquid height, H1 =	2.5 m

The seismic base shear is the combination of the impulsive and convective components:

V = Vi + Vc

$$V_i = \frac{S_{ai} W_i}{(R/I)} \qquad V_c = \frac{S_{ac} I}{1.5} W_c$$

Conservatively considering Sai = Sds, and R = 1, I = 1.15

Hence, Vi = 28.51 kN

Natural time period for first convective mode of sloshing :

$$T_c = 2\pi \sqrt{\frac{D}{3.68\, g\, tan\, h\left(\frac{3.68H}{D}\right)}} = 3.0792 \text{ sec}$$

For $T_c > T_L$:

$$S_{ac} = \frac{1.5\, S_{D1} T_L}{T_c^2} = 0.00823$$

For $T_c \le T_L$:

$$S_{ac} = \frac{1.5\, S_{D1}}{T_c} = 0.006333$$

Thus, Vc = 1.61 kN

Hence, seismic base shear, V = 30.12 kN

Moment at base due to seismic = 30.12x3.6x0.5= 54.22 kN.m

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5.3 DESIGN OF FOUNDATION TYPE F4 FOR 3660-TNK-005

	Along-X	Along-Z
Size of column :	0.60m	0.60m
Size of fdn. raft :	2.00m	2.00m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.55 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	20 kN/m^3	

SYMBOLS AND NOTATIONS

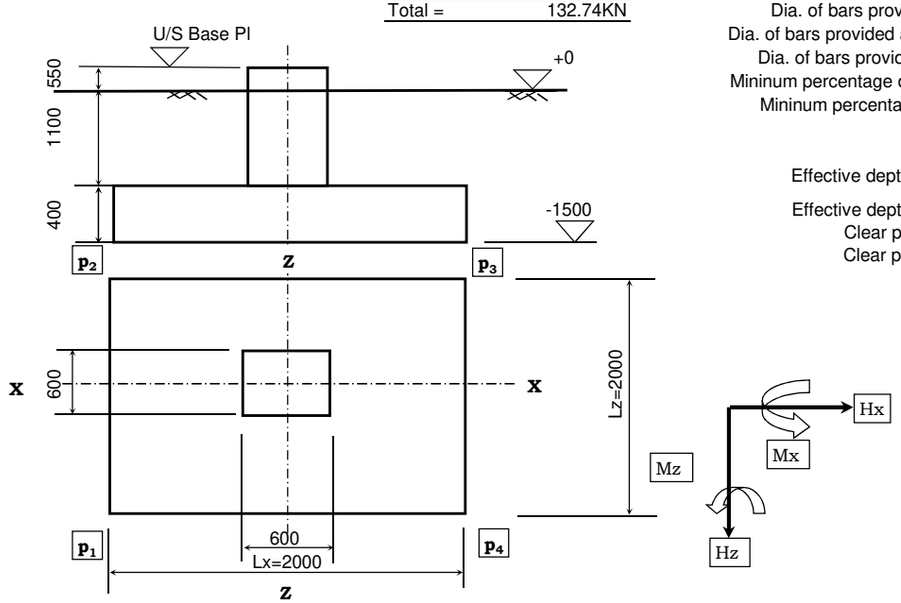
Dead Load	D
Seismic Load	E
Wind Load	W
Live Load	L
Fluid Load	F

Calculation of Weight of Foundation :

Weight of raft & Col.= (2*2*0.4*24)+(0.6*0.6*1.65*24) = 52.66KN

Weight of soil = ((2*2)-(0.6*0.6))*1.1*20 = 80.08KN

Total = 132.74KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.7m

Clear projection along X direction = 0.7m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	14.00	0.00	0.00	0.00	0.00
2	Seismic Load (E)	82.50	50.40	0.40	15.40	1.40
3	Wind Load (W)	20.00	4.30	0.30	1.50	0.30
4	Live Load (L)	0.00	0.00	0.00	0.00	0.00
5	Fluid Load (F)	74.10	49.10	0.10	14.80	0.10

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D + F	88.1	49.1	14.8	0.1	0.1	220.84	49.10	115.46	0.10	0.31
202	D + F + L	88.1	49.1	14.8	0.1	0.1	220.84	49.10	115.46	0.10	0.31
203	D + F + W	108.1	53.4	16.3	0.4	0.4	240.84	53.40	125.77	0.40	1.22
204	0.6 D + W	28.4	4.3	1.5	0.3	0.3	161.14	4.30	10.32	0.30	0.92
205	0.6 D + 0.7 E	66.2	35.3	10.8	0.3	1.0	145.79	35.28	83.10	0.28	1.55
Ultimate Conditions											
301	1.4 D + 1. 4F	123.3	68.7	20.7	0.1	0.1	309.18	68.74	161.64	0.14	0.43
302	1.2 D + 1.2 F + 1.6 L	105.7	58.9	17.8	0.1	0.1	291.56	58.92	138.55	0.12	0.37
303	1.2 D + 1.0 E + 0.5 L	99.3	50.4	15.4	0.4	1.4	258.59	50.40	118.72	0.40	2.22
304	0.9 D + 1.0 E	95.1	50.4	15.4	0.4	1.4	214.57	50.40	118.72	0.40	2.22
305	0.9 D + 1.6 W	44.6	6.9	2.4	0.5	0.5	164.07	6.88	16.50	0.48	1.46
306	1.2 D + 1.6 W + 0.5 L	48.8	6.9	2.4	0.5	0.5	208.09	6.88	16.50	0.48	1.46

Check for overturning and sliding:

	Load Cases:	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F	220.84	49.10	115.46	0.10	0.31	-	-	-	-
202	D + F + L	220.84	49.10	115.46	0.10	0.31	-	-	-	-
203	D + F + W	240.84	53.40	125.77	0.40	1.22	1.91	197.41	1.80	240.84
204	0.6 D + W	161.14	4.30	10.32	0.30	0.92	15.62	176.11	14.99	214.85
205	0.6 D + 0.7 E	145.79	35.28	83.10	0.28	1.55	1.75	93.82	1.65	208.28

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Load Combination & Calculation of Base pressure

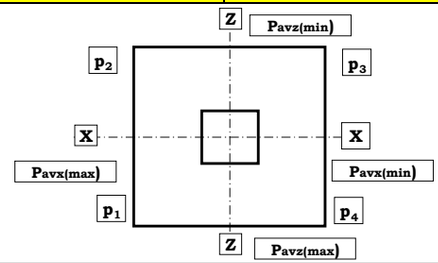
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m²

Net SBC of soil for Dead Load = 350 kN/m² For Seismic load case = 437.5 kN/m²

Gross SBC of soil for Dead Load = 380 kN/m² For Seismic load case = 467.5 kN/m²



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m²)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D + F	220.84	115.46	0.31	55.21	86.59	0.23	142.03	141.57	0.00	0.00	112.03	141.80	0.00	71.02	70.79	O.K
202	D + F + L	220.84	115.46	0.31	55.21	86.59	0.23	142.03	141.57	0.00	0.00	112.03	141.80	0.00	71.02	70.79	O.K
203	D + F + W	240.84	125.77	1.22	60.21	94.33	0.92	155.45	153.62	0.00	0.00	125.45	154.54	0.00	77.73	76.81	O.K
204	0.6 D + W	161.14	10.32	0.92	40.29	7.74	0.69	48.71	47.34	33.24	31.86	18.71	48.03	32.55	40.98	39.60	O.K
205	0.6 D + 0.7 E	145.79	83.10	1.55	36.45	62.33	1.17	99.94	97.61	0.00	0.00	69.94	98.78	0.00	49.97	48.81	O.K
301	1.4 D + 1. 4F	309.18	161.64	0.43	77.29	121.23	0.32	198.84	198.20	0.00	0.00	168.84	198.52	0.00	99.42	99.10	O.K
302	1.2 D + 1.2 F + 1.6 L	291.56	138.55	0.37	72.89	103.91	0.27	177.07	176.52	0.00	0.00	147.07	176.80	0.00	88.54	88.26	O.K
303	1.2 D + 1.0 E + 0.5 L	258.59	118.72	2.22	64.65	89.04	1.67	155.35	152.02	0.00	0.00	125.35	153.69	0.00	77.68	76.01	O.K
304	0.9 D + 1.0 E	214.57	118.72	2.22	53.64	89.04	1.67	144.35	141.02	0.00	0.00	114.35	142.69	0.00	72.18	70.51	O.K
305	0.9 D + 1.6 W	164.07	16.50	1.46	41.02	12.38	1.10	54.49	52.30	29.74	27.54	24.49	53.40	28.64	42.12	39.92	O.K
306	1.2 D + 1.6 W + 0.5 L	208.09	16.50	1.46	52.02	12.38	1.10	65.50	63.30	40.74	38.55	35.50	64.40	39.65	53.12	50.93	O.K

Max :	309	162	2	Max pressure at base of foundation without surcharge excluding weight of soil (Dead load cases) =	112.03	kN/m2
Min :	146	10	0	Max pressure at base of foundation including surcharge excluding weight of soil (Dead load cases) =	132.03	kN/m2
				Max pressure at base of foundation without surcharge excluding weight of soil (Dead+Temperature+Seismic load cases) =	168.84	kN/m2
				Max pressure at base of foundation including surcharge excluding weight of soil (Dead+ Temperature+Seismic load cases) =	188.84	kN/m2

Sample calculation of values of base pressure:

Load case : D + F + L

Maximum pressure, p_1 (pmax) =	55.21+86.59+0.23=	142.03 kN/m²	Maximum average Pressure along X, $p_{avx(max)}$ =	(142.03+141.57)/2= 141.80 kN/m2
p_2=	55.21+86.59-0.23=	141.57 kN/m²	Minimum average Pressure along X, $p_{avx(minx)}$ =	(0.00+0.00)/2= -31.38 kN/m2
p_3=	55.21-(86.59+0.23)=	-31.61 kN/m²	Maximum average Pressure along Z, $p_{avz(max)}$ =	(142.03+0.00)/2= 55.44 kN/m2
p_4=	55.21-86.59+0.23=	-31.15 kN/m²	Minimum average Pressure along Z, $p_{avz(minx)}$ =	(0.00+141.57)/2= 54.98 kN/m2
Maxm pressure excluding weight of soil (p_{max}-γh)=	142.03-1.5*20=	112.03 kN/m²		

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

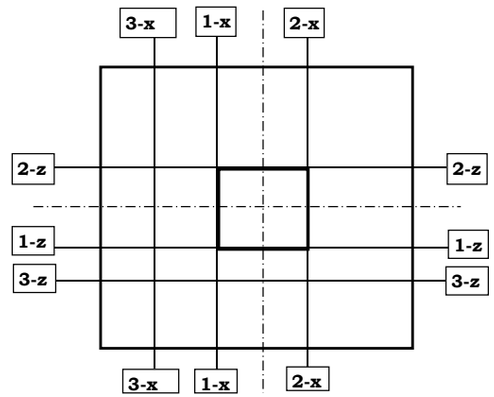
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.326)+(0.6+2*0.5*0.326))*2

= 3.704 m

Punching Area= ((0.6+2*0.5*0.326)*(0.6+2*0.5*0.326))

= 0.86 m2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.4*24+1.1*20)

= 31.60 kN/m2

Area of footing, Af= 4.00 m2

Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	$B.M_{1x}$ kNm	$B.M_{2x}$ kNm	$S.F_{3x}$ kN	S.F(punching) kN
Ultimate Conditions								
301	1.4 D + 1. 4F	129.04	69.48	162.98	35.22	-2.07	53.40	143.48
302	1.2 D + 1.2 F + 1.6 L	114.92	61.88	145.15	30.52	-2.69	46.32	129.65
303	1.2 D + 1.0 E + 0.5 L	99.90	53.79	126.18	25.52	-3.35	38.78	103.77
304	0.9 D + 1.0 E	92.75	49.94	117.15	23.14	-3.66	35.20	69.21
305	0.9 D + 1.6 W	44.73	37.31	48.97	4.63	-0.02	7.01	29.57
306	1.2 D + 1.6 W + 0.5 L	55.74	48.31	59.97	7.33	2.68	10.95	64.13

Sample calculation of BM & SF

Load case : 301

p_{1x} = 198.52-(198.52-0)*0.7/2 = 129.04 kN/m^2

p_{2x} = 0+(198.52-0)*0.7/2 = 69.48 kN/m^2

p_{3x} = 198.52-(198.52-0)*(0.7-0.342)/2 = 162.98 kN/m^2

Overhang = 0.7 m

Effective depth = 0.342 m

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BM_{1x} = (129.04-31.6)*0.7*0.7/2+0.5*(198.52-129.04)*0.7*0.7*0.67 = 35.22 kN-m/m

BM_{2x} = (69.48-31.6)*0.7*0.7/2-0.5*(-0+69.48)*0.7*0.7*0.67 = -2.07 kN-m/m

SF_{3x} = (162.98-31.6)*(0.7-0.342)+0.5*(198.52-162.98)*(0.7-0.342) = 53.40 kN/m

$S.F_{(punching)}$ = (4-0.86)*(77.29-31.6) = 143.48 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	35.22	0.30	0.40	0.001	329.17	615.60	16ϕT	326.61	300	200	1005.31
At top	-3.66	-0.03	0.04	0.000	34.03	205.20	16ϕT	979.83	300	200	1005.31

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 53.40 kN/m

Shear stress , τ_v= (53.4*1000)/(1000*342)

= 0.16 N/mm2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 143.48 kN

Strength of Concrete in shear Vc1 = 3747.15 kN Table 22.6.5.2_ (a)

Strength of Concrete in shear Vc2 = 2402.33 kN Table 22.6.5.2_ (b)

Strength of Concrete in shear Vc3 = 2498.10 kN Table 22.6.5.2_ (c)

Design Vc= 2402.33 kN

Allowable shear stress, $(\phi\ Vc)/bd$ = 1.42 N/mm2

Design punching shear stress, τ = (143.48*1000)/(3704*342)

= 0.11 N/mm2 **No shear reinforcement reqd.**

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Calculation of Bending Moment and Shear Force in Z direction :

p_{1z} =pressure at the face of column at section 1z-1z in kN/m2.
p_{2z} =pressure at the face of column at section 2z-2z in kN/m2.
p_{3z} =pressure at a distance d from the face of column at section 3z-3z in kN/m2.
$B.M_{1z}$ =Moment at the face of column at section 1z-1z in kN-m/m.
$B.M_{2z}$ =Moment at the face of column at section 2z-2z in kN-m/m.
$S.F_{3z}$ = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
301	1.4 D + 1. 4F	99.31	99.21	99.36	16.61	16.55	25.35
302	1.2 D + 1.2 F + 1.6 L	88.44	88.36	88.49	13.94	13.89	21.29
303	1.2 D + 1.0 E + 0.5 L	77.10	76.59	77.37	11.24	10.93	17.18
304	0.9 D + 1.0 E	71.60	71.09	71.87	9.89	9.58	15.12
305	0.9 D + 1.6 W	41.35	40.69	41.71	2.52	2.10	3.86
306	1.2 D + 1.6 W + 0.5 L	52.35	51.70	52.71	5.21	4.80	7.97

Sample calculation of BM & SF

Load case : 301

p_{1z} = 99.42-(99.42-99.1)*0.7/2= 99.31 kN/m2

p_{2z} = 99.1+(99.42-99.1)*0.7/2= 99.21 kN/m2 Overhang = 0.7 m

p_{3z} = 99.42-(99.42-99.1)*(0.7-0.326)/2= 99.36 kN/m2 Effective depth = 0.326 m

BM_{1z} = (99.31-31.6)*0.7*0.7/2+0.5*(99.42-99.31)*0.7*0.7*0.67 = 16.61 kN-m/m

BM_{2z} = (99.21-31.6)*0.7*0.7/2-0.5*(-99.1+99.21)*0.7*0.7*0.67 = 16.55 kN-m/m

SF_{3z} = (99.36-31.6)*(0.7-0.326)+0.5*(99.42-99.36)*(0.7-0.326) = 25.35 kN/m

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Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm2)	ρ=	Ast (mm2)	Minm. Ast (mm2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm2)
At bottom	16.61	0.156	0.208	0.0005	170.25	586.80	16ϕT	342.64	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	195.60	10ϕT	401.53	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.74 N/mm^2

Shear at a distance 'd' from face of column, V_u= 25.35 kN/m

Shear stress , τ_v= (25.35*1000)/(1000*326)

= 0.08 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	16ϕT	200	c/c
Along Z Direction :				
	At Bottom	16ϕT	200	c/c
	At Top	10ϕT	200	c/c

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5.4 DESIGN OF FOOTING F5 for 3660-TNK-005

Size of Columns at node	Along-X	Along-Z
1	0.60m	0.60m
2	0.60m	0.60m
Size of fdn. raft:	4.50m	1.50m
Thickness of raft:	0.60	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.55	m
U/S level of fdn:	-1.50	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

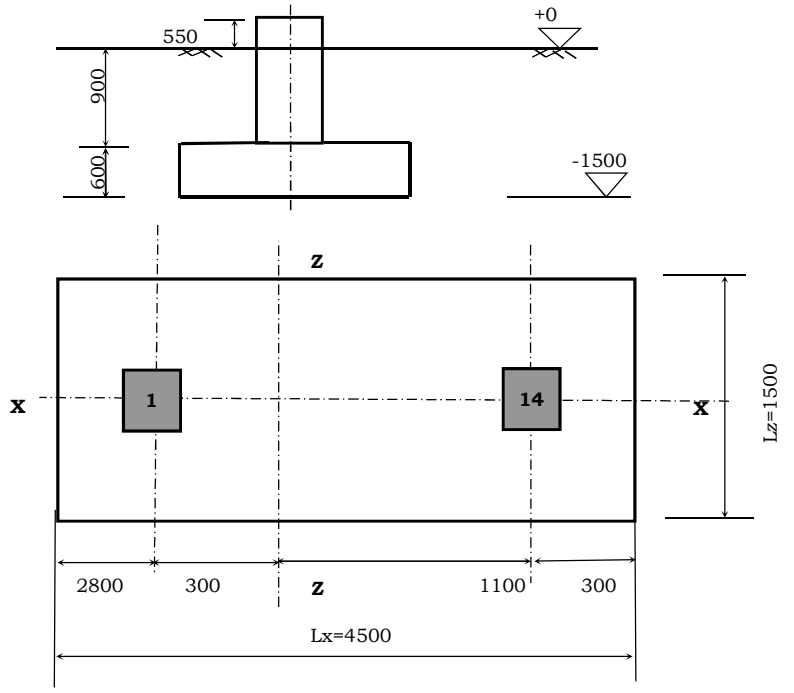
Calculation of Weight of Foundation :

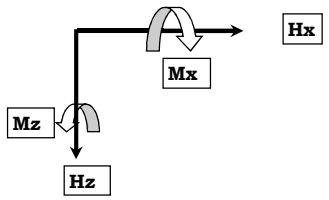
Weight of raft = (4.5*1.5*0.6*25) = 101.25KN

Weight of soil = ((4.5*1.5)-2*(0.6*0.6))*0.9*20 = 108.54KN

Total = 209.79KN

Submerged wt. = 209.79-4.5*1.5*1.5*10 = 108.54KN





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing

Load at bottom of fooring	Node 1					Node 14				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1. DL	136.5	-18.7	40.7	-24.5	-59.9	135.3	18.8	-41.0	-24.7	-60.1
202 1 DL + 1 WL	153.5	-20.6	43.3	-26.7	-65.2	161.5	12.4	-29.4	-27.3	-66.4
203 1 DL + 0.7 EL	221.9	-33.8	69.2	-44.0	-107.4	218.6	5.9	-16.2	-44.4	-107.9
204 0.6 DL + 1 WL	98.9	-13.1	27.0	-16.9	-41.3	107.4	4.9	-13.0	-17.4	-42.3
205 0.6 DL + 0.7 EL	167.3	-26.3	52.9	-34.2	-83.4	164.5	-1.7	0.2	-34.5	-83.8

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Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1. DL	481.6	0.10	-1.6	-49.2	-120.1	666.8	3.0	1926.3	3.9
202 1 DL + 1 WL	524.8	-8.20	22.7	-54.0	-131.6	52.0	3.0	25.6	3.9
203 1 DL + 0.7 EL	650.3	-27.97	49.3	-88.4	-215.3	29.7	2.3	9.3	2.9
204 0.6 DL + 1 WL	416.1	-8.24	23.4	-34.3	-83.6	40.1	3.7	20.2	4.8
205 0.6 DL + 0.7 EL	541.6	-28.01	50.0	-68.7	-167.2	24.4	2.4	7.7	3.2
					Min.=	**24.4**	**2.3**	**7.7**	**2.9**

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Base pressure check of footing

	Node	L/C	Horizontal Fx N/mm2	Vertical Fy N/mm2	Horizontal Fz N/mm2
Max Px	495	201	0	0.05	0
Min Px	495	201	0	0.05	0
Max Py	501	203	0	0.21	0
Min Py	502	203	0	-0.06	0
Max Pz	495	201	0	0.05	0
Min Pz	495	201	0	0.05	0

Gross Maximum Base pressure, p_{max} = 211.0 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.8 m
Unit weight of soil = 18 KN/m^3

Gross Safe Bearing Capacity = 464.4 KN/m^2
> 211.0 KN/m^2
Hence OK

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Design of foundation raft F8 & F9

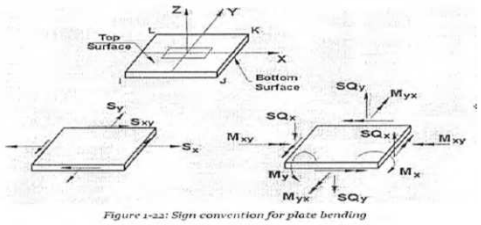
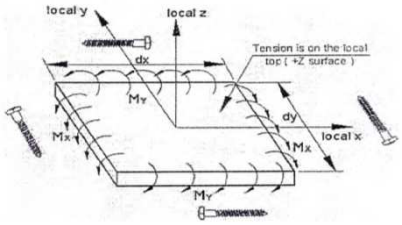


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F8&F9_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	2151	0.349	-0.121	0	0	0	3.123	31.462	-25.127	28.25	56.589	103 1.2 DL + 1.0 EL
Min Qx	2020	-0.475	-0.114	0	0	0	57.721	32.594	0	57.721	32.594	103 1.2 DL + 1.0 EL
Max Qy	2004	0.001	0.174	0	0	0	-0.354	33.168	3.808	-4.162	36.976	103 1.2 DL + 1.0 EL
Min Qy	2464	-0.057	-0.522	0	0	0	34.952	72.547	23.53	58.482	96.077	103 1.2 DL + 1.0 EL
Max Sx	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sx	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Sy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Sxy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sxy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Mx	2020	-0.475	-0.114	0	0	0	57.721	32.594	31.196	88.917	63.79	103 1.2 DL + 1.0 EL
Min Mx	3034	0.021	-0.013	0	0	0	-59.83	-0.32	-1.742	-61.575	-2.062	103 1.2 DL + 1.0 EL
Max My	2464	-0.057	-0.522	0	0	0	34.952	72.547	23.53	58.482	96.077	103 1.2 DL + 1.0 EL
Min My	1293	0.002	-0.118	0	0	0	34.205	-49.386	0.259	34.464	-49.645	103 1.2 DL + 1.0 EL
Max Mxy	2200	-0.441	-0.223	0	0	0	50.357	36.499	33.304	83.661	69.803	103 1.2 DL + 1.0 EL
Min Mxy	1344	0.001	-0.058	0	0	0	-3.654	-21.812	-36.924	-40.578	-58.736	103 1.2 DL + 1.0 EL

PLATE CENTER STRESS : FDN_F8&F9_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	2151	0.349	-0.121	0	0	0	3.123	31.462	-25.127	28.25	56.589	103 1.2 DL + 1.0 EL
Min Qx	2020	-0.475	-0.114	0	0	0	57.721	32.594	0	57.721	32.594	103 1.2 DL + 1.0 EL
Max Qy	2004	0.001	0.174	0	0	0	-0.354	33.168	3.808	-4.162	36.976	103 1.2 DL + 1.0 EL
Min Qy	2464	-0.057	-0.522	0	0	0	34.952	72.547	23.53	58.482	96.077	103 1.2 DL + 1.0 EL
Max Sx	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sx	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Sy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Sxy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Min Sxy	564	0.008	-0.01	0	0	0	0.075	-0.089	0.008	0.083	-0.097	101 1.4 DL
Max Mx	2020	-0.475	-0.114	0	0	0	57.721	32.594	31.196	88.917	63.79	103 1.2 DL + 1.0 EL
Min Mx	3034	0.021	-0.013	0	0	0	-59.83	-0.32	-1.742	-61.575	-2.062	103 1.2 DL + 1.0 EL
Max My	2464	-0.057	-0.522	0	0	0	34.952	72.547	23.53	58.482	96.077	103 1.2 DL + 1.0 EL
Min My	1293	0.002	-0.118	0	0	0	34.205	-49.386	0.259	34.464	-49.645	103 1.2 DL + 1.0 EL
Max Mxy	2200	-0.441	-0.223	0	0	0	50.357	36.499	33.304	83.661	69.803	103 1.2 DL + 1.0 EL
Min Mxy	1344	0.001	-0.058	0	0	0	-3.654	-21.812	-36.924	-40.578	-58.736	103 1.2 DL + 1.0 EL

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Material Property :

Grade of concrete =		fc'28	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.85
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

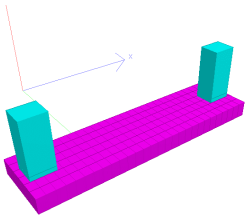
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2 D	88.92	top face	50	542	88.917	0.40	0.0010	525.3	975.6	16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	-61.58	bottom face	50	542	61.575	0.28	0.0007	362.8		16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	83.66	top face	50	542	83.661	0.38	0.0009	494.0		16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	-40.58	top face	50	542	40.578	0.18	0.0004	238.6		16	206.17	200	1005.8	17.7	20.83	0.075	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.349	1	209.4	28.25	0.002	0.542	527.573	475.978	831.718	475.98	1777	> 209.4	ok	357	φVc>Vu	not required
	Min Qx	0.475	1	285	57.721			504.638			475.98	1777	> 285	ok	357	φVc>Vu	not required

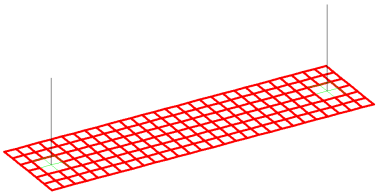
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
103 1.2 D	96.08	top face	50	542	96.077	0.44	0.0010	568.0	975.6	16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	-49.65	bottom face	50	542	49.645	0.23	0.0005	292.2		16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	69.80	bottom face	50	542	69.803	0.32	0.0008	411.6		16	206.17	200	1005.8	17.7	20.83	0.075	tension control
103 1.2 D	-58.74	top face	50	542	58.736	0.27	0.0006	346.0		16	206.17	200	1005.8	17.7	20.83	0.075	tension control

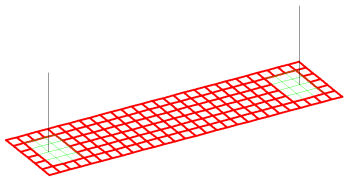
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	103 1.2 D	0.174	1	104.4	36.976	0.002	0.542	485.045	475.978	fc'831.7	475.98	499	> 104.4	ok	357	φVc>Vu	not required
	103 1.2 D	0.522	1	313.2	96.077			489.09			475.98	499	> 313.2	ok	357	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

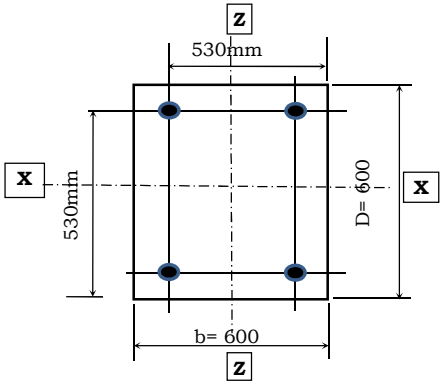
Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c

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5.5 DESIGN OF COLUMN FOR 3660-TNK-005



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1 of SBC-304
Dia. of main reinf.=	20mm	
Dia. of link=	10mm	
Effective cover =	70mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ = 0.65 Table 21.2.2 of SBC-304

Unsupproted length of column about X (l_{ux}) =	1650mm
Unsupproted length of column about Z (l_{uz}) =	1650mm

For Columns not braced against sidesway, Where, Effective length factor K= 2

Slenderness ratio, $K\,l_{ux} / r_x$ = 18.33 < 22 SECTION 6.2.5a Radius of gyration r_x= 180 mm

$K\,l_{uz} / r_z$ = 18.33 < 22 SECTION 6.2.5a Radius of gyration r_z= 180 mm

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	14.00	0.00	0.00	0.00	0.00
2	Seismic Load (E)	82.50	50.40	0.40	15.40	1.40
3	Wind Load (W)	20.00	4.30	0.30	1.50	0.30
4	Live Load (L)	0.00	0.00	0.00	0.00	0.00
5	Fluid Load (F)	74.10	49.10	0.10	14.80	0.10
			γ_x =	0.80		
			γ_z =	0.80		

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Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'cbD (N/mm2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
301	1.4 D + 1.4 F	12.33	13.41	0.04	18.98	20.64	0.06	0.02
302	1.2 D + 1.2 F + 1.6 L	10.57	11.50	0.03	16.26	17.69	0.05	0.01
303	1.2 D + 1.0 E + 0.5 L	9.93	9.86	0.21	15.28	15.16	0.32	0.01
304	0.9 D + 1.0 E	9.51	9.86	0.21	14.63	15.16	0.32	0.01
305	0.9 D + 1.6 W	4.46	1.38	0.13	6.86	2.12	0.20	0.01
306	1.2 D + 1.6 W + 0.5 L	4.88	1.38	0.13	7.51	2.12	0.20	0.01

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	Asc pvd. = 3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total Asc = 1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_c b^2 D$	$R_{nx}=M_{nox}/f'_c b^2 D$						
301	0.04	0.04	30.2	30.2	0.55	1.16	0.64	O.K
302	0.04	1.04	30.2	786.2	0.55	1.16	0.54	O.K
303	0.04	2.04	30.2	1542.2	0.55	1.16	0.45	O.K
304	0.04	3.04	30.2	2298.2	0.55	1.16	0.45	O.K
305	0.04	4.04	30.2	3054.2	0.55	1.16	0.05	O.K
306	0.04	5.04	30.2	3810.2	0.55	1.16	0.05	O.K

Links, 10mm dia @ 300 C/C

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REFERENCES:

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) clause 4.1.

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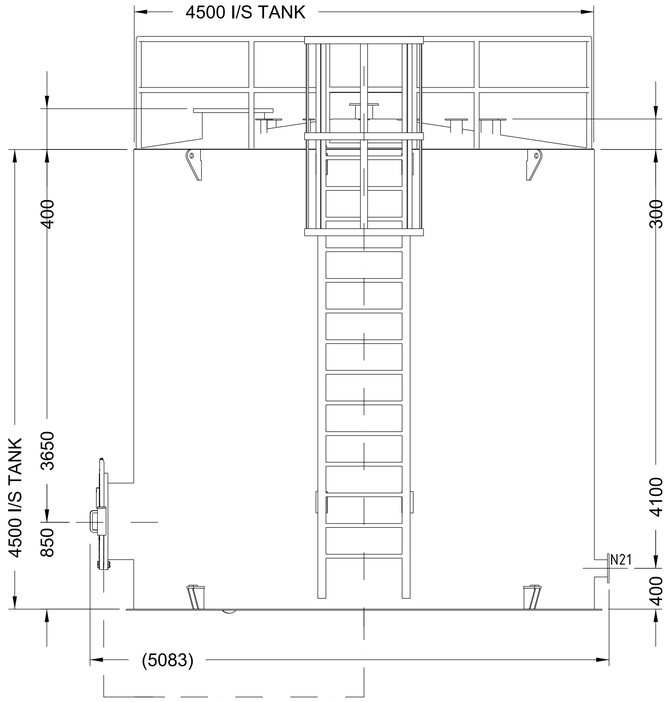
ATTACHMENT 1 EQUIPMENT DRAWING

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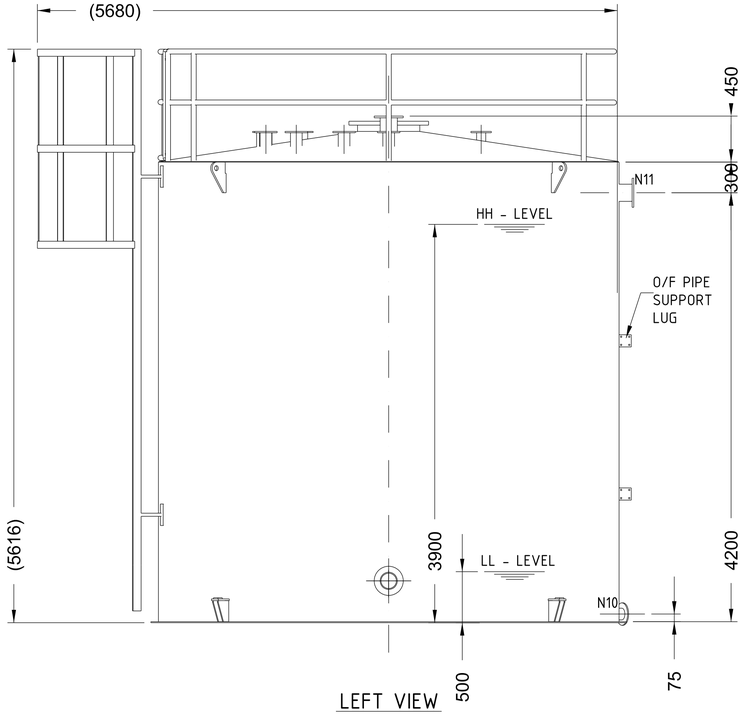
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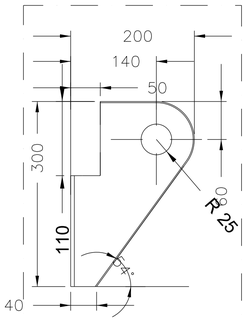
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REV: A01



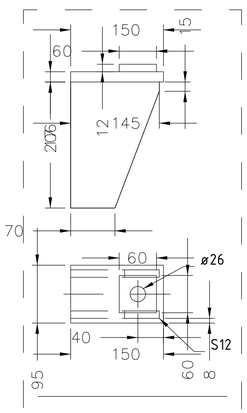
FRONT VIEW



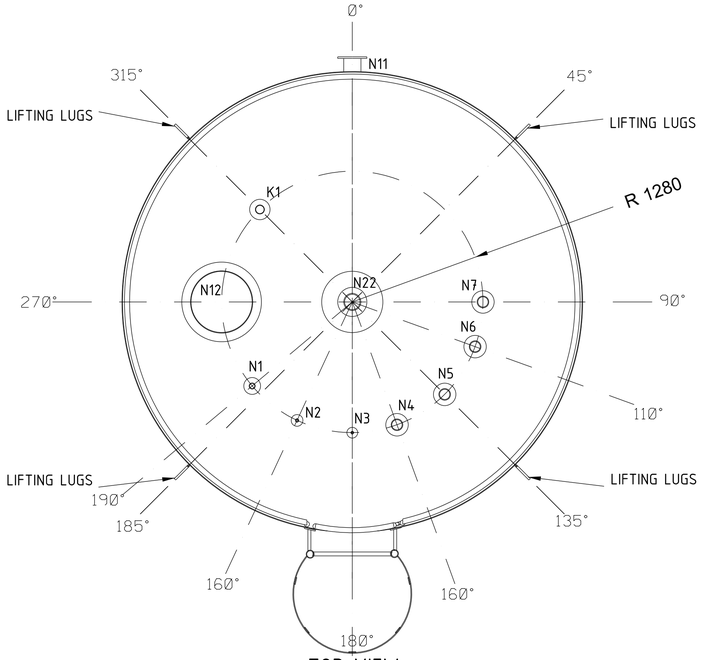
LEFT VIEW



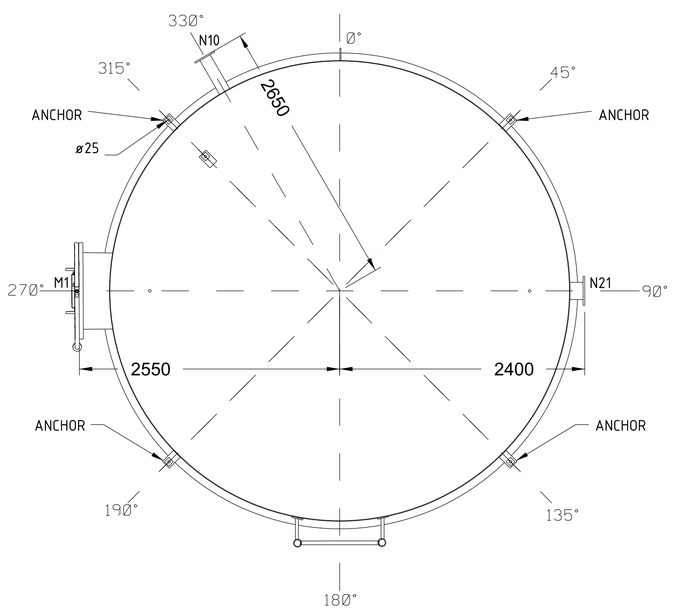
LIFTING LUGS
NOT IN SCALE



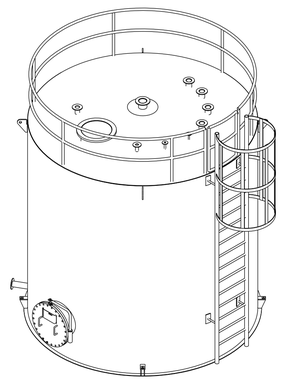
ANCHORING SADDLE
NOT IN SCALE



TOP VIEW



ACHORING



ISOMETRIC VIEW

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3660-TNK-010
DESCRIPTION	LG ELUTION TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	6 mm
BOTTOM THICKNESS	10 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4200mmx4820mmx5050mm
SHELL HEIGHT	3950mm
CAPACITY: TOTAL VOLUME	67m³
CAPACITY: LIVE VOLUME	55m³
SEISMIC DESIGN /CODE	Site B / sBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	76000 kg
TOTAL WEIGHT (EMPTY)	7000 kg
RUBBER LINING WEIGHT	NO LINING
TANK WEIGHT	7000 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	2"	150	190°	300	1280	4500	ASME B16.5	INLET
N2	1"	150	160°	300	1280	4500	ASME B16.5	INLET
N3	3/4"	150	180°	300	1280	4500	ASME B16.5	INLET
N4	4"	150	160°	300	1280	4500	ASME B16.5	INLET
N5	4"	150	135°	300	1280	4500	ASME B16.5	INLET
N6	4"	150	110°	300	1280	4500	ASME B16.5	INLET
N21	6"	150	90°	150	2250	400	ASME B16.5	OUTLET
N22	6"	150	-	450	0	4500	ASME B16.5	VENT
N10	4"	150	330°	400	2250	75	ASME B16.5	DRAIN
N11	6"	150	0°	150	2250	4200	ASME B16.5	EMERGENCY OVERFLOW
N12	24"	150	270°	400	1280	4500	API 650	INSPECTION HOLE
M1	24"	150	270°	300	2250	850	API 650	MANHOLE
K1	4"	150	315°	300	1280	4500	ASME B16.5	LEVEL INDICATOR
N7	4"	150	90°	300	1280	4500	ASME B16.5	SPARE

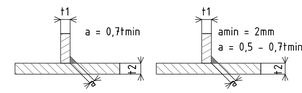
NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. EQUIPMENT INSULATION FOR HEAT RESERVATION

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	01-07-20	FIRST ISSUE	KL	JS	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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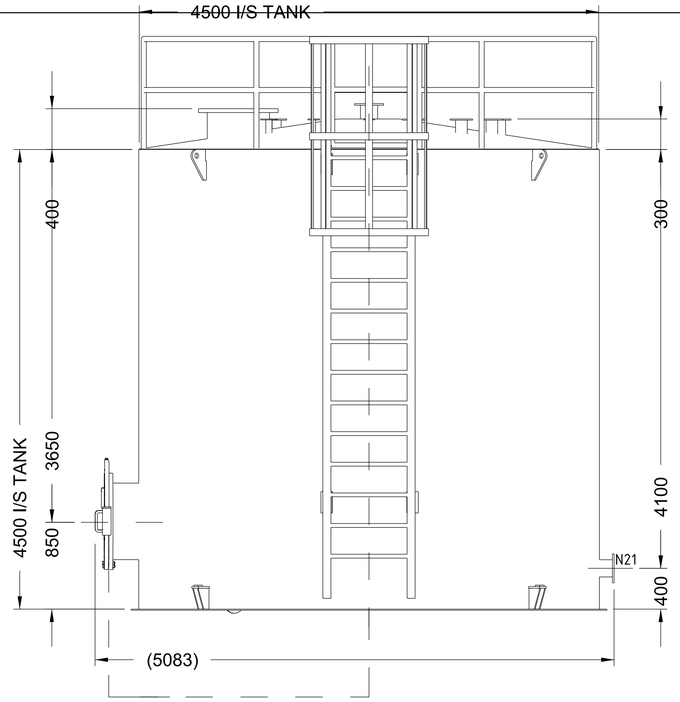
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OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
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DRAWN	KimLop	01-07-20
CHECKED	JonSal	01-07-20
ENGINEER	SouCha	01-07-20
APPROVED	RicLeh	01-07-20
EMPLOYER		
SCALE	1:30	A1

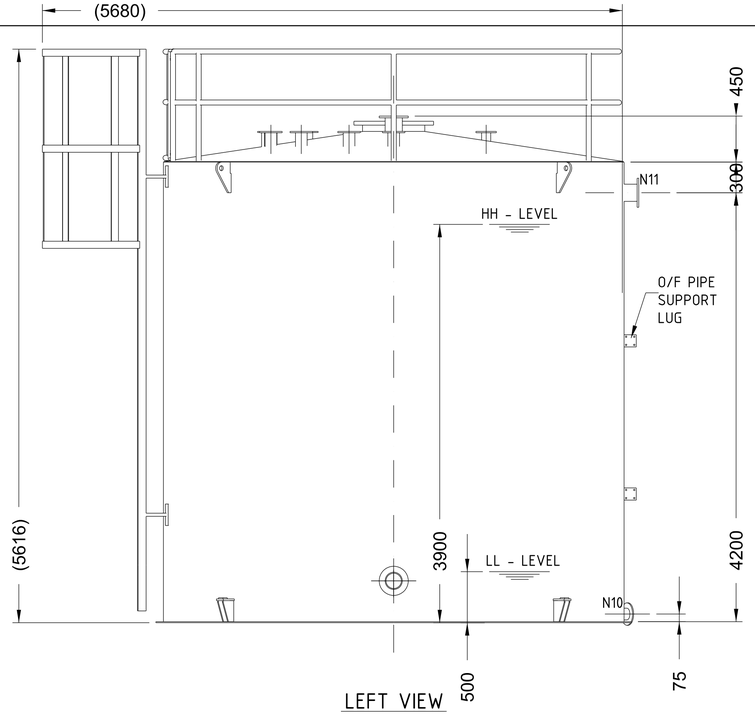
MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
LOW GRADE ACID WASH, ELUTION AND CARBON REGENERATION
3660-TNK-010 COPPER ELUATE TANK

DRAWING NUMBER: MD-216-3660-EG-ME-GAD-1004
REV: A01

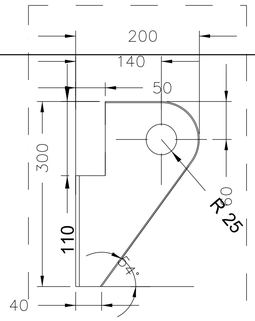
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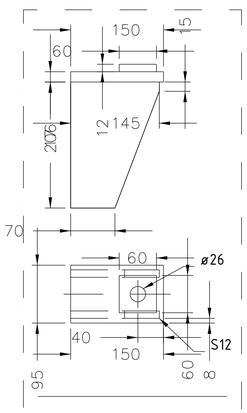
FRONT VIEW



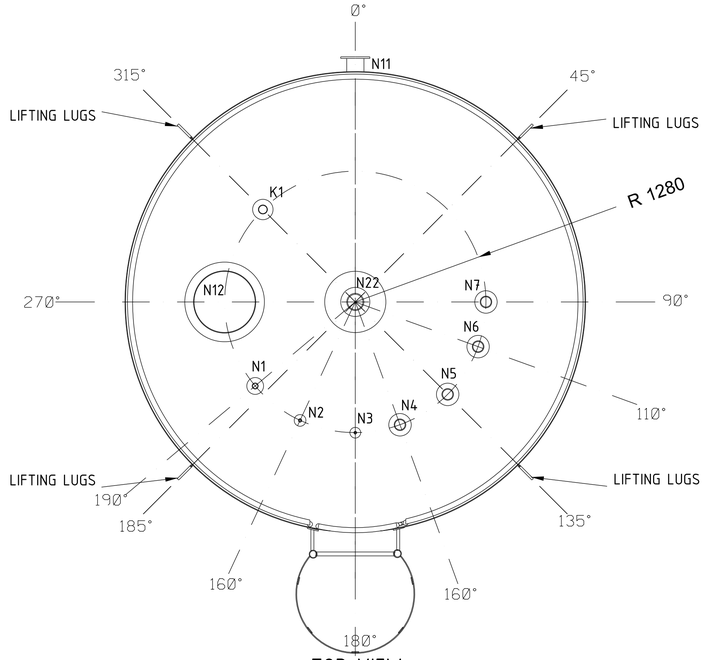
LEFT VIEW



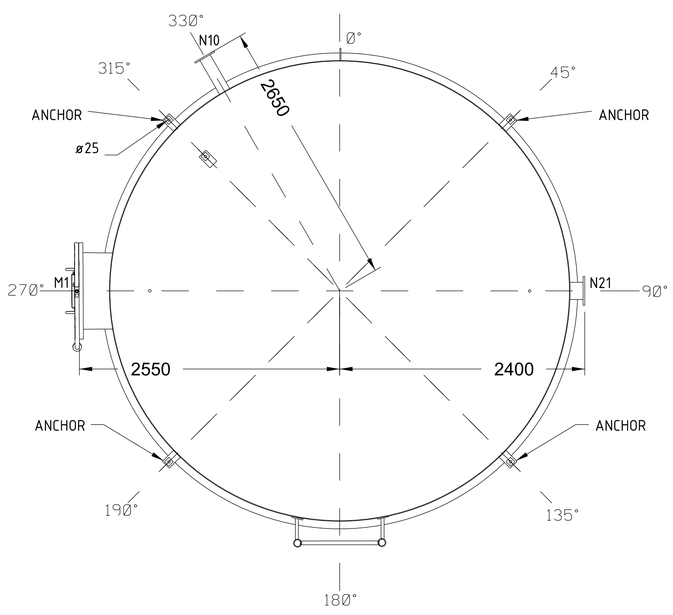
LIFTING LUGS
NOT IN SCALE



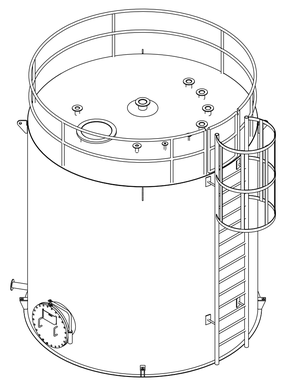
ANCHORING SADDLE
NOT IN SCALE



TOP VIEW



ACHORING



ISOMETRIC VIEW

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3660-TNK-002
~~DESCRIPTION~~	~~LG ELUTION TANK~~
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	6 mm
BOTTOM THICKNESS	10 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4200mmx4820mmx5050mm
SHELL HEIGHT	3950mm
CAPACITY: TOTAL VOLUME	67m³
CAPACITY: LIVE VOLUME	55m³
SEISMIC DESIGN /CODE	Site B / sBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	76000 kg
TOTAL WEIGHT (EMPTY)	7000 kg
RUBBER LINING WEIGHT	NO LINING
TANK WEIGHT	7000 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	2"	150	190°	300	1280	4500	ASME B16.5	INLET
N2	1"	150	160°	300	1280	4500	ASME B16.5	INLET
N3	3/4"	150	180°	300	1280	4500	ASME B16.5	INLET
N4	4"	150	160°	300	1280	4500	ASME B16.5	INLET
N5	4"	150	135°	300	1280	4500	ASME B16.5	INLET
N6	4"	150	110°	300	1280	4500	ASME B16.5	INLET
N21	6"	150	90°	150	2250	400	ASME B16.5	OUTLET
N22	6"	150	-	450	0	4500	ASME B16.5	VENT
N10	4"	150	330°	400	2250	75	ASME B16.5	DRAIN
N11	6"	150	0°	150	2250	4200	ASME B16.5	EMERGENCY OVERFLOW
N12	24"	150	270°	400	1280	4500	API 650	INSPECTION HOLE
M1	24"	150	270°	300	2250	850	API 650	MANHOLE
K1	4"	150	315°	300	1280	4500	ASME B16.5	LEVEL INDICATOR
N7	4"	150	90°	300	1280	4500	ASME B16.5	SPARE

NOTE:

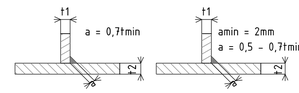
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.
4. EQUIPMENT INSULATION FOR HEAT RESERVATION

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	01-07-20	FIRST ISSUE	KL	JS	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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DRAWN	KimLop	01-07-20	
CHECKED	JonSal	01-07-20	
ENGINEER	SouCha	01-07-20	
APPROVED	RicLeh	01-07-20	
EMPLOYER			
SCALE	1:30	A1	

LARSEN & TOUBRO DRAWING NUMBER 019045_450_M_G_02918_045 — REVISION 0

OUTOTEC DRAWING NUMBER 900707_GAU01_EE13_MTA01_36607 — DOCUMENT ID OU100680641 — REVISION 0

MANSOURAH - MASSARAH GOLD PROJECT

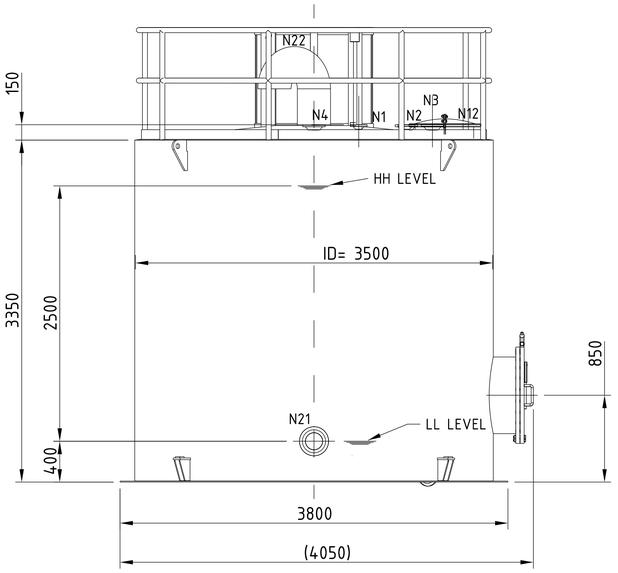
OUTLINE DRAWING

LOW GRADE ACID WASH, ELUTION AND CARBON REGENERATION

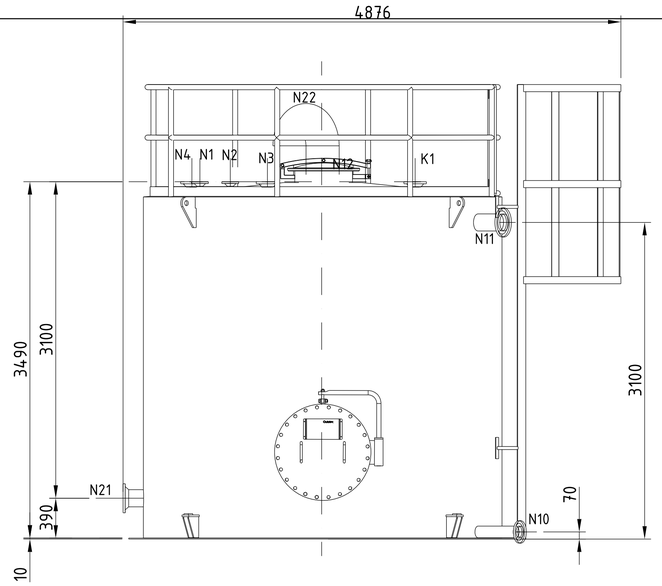
3660-TNK-002 LG ELUTION TANK

DRAWING NUMBER MD-216-3660-EG-ME-GAD-1035 — REV A01

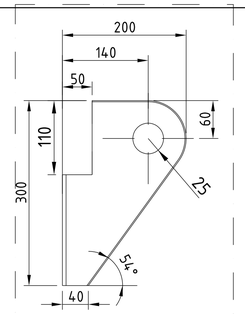
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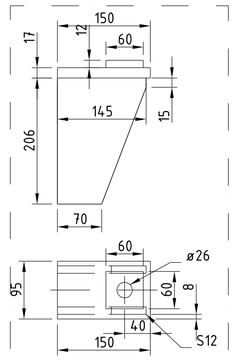
FRONT VIEW



RIGHT VIEW

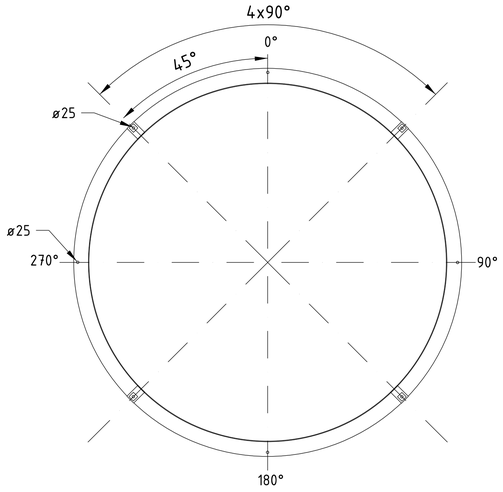


LIFTING LUGS
NOT IN SCALE

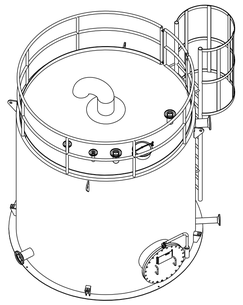


ANCHORING SADDLE
NOT IN SCALE

TOP VIEW



ACHORING



ISOMETRIC VIEW

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3660-TNK-001
DESCRIPTION	LG ACID WASH TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	6 mm
BOTTOM THICKNESS	10 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm (6 THK. RUBBER LINING PROVIDED)
CORROSION ALLOWANCE: BOTTOM	2 mm (6 THK. RUBBER LINING PROVIDED)
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4875mmx4050mmx4450mm
SHELL HEIGHT	3350mm
CAPACITY: TOTAL VOLUME	31m³
CAPACITY: LIVE VOLUME	25m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	36940 kg
TOTAL WEIGHT (EMPTY)	3840 kg
RUBBER LINING WEIGHT	340 kg
TANK WEIGHT	3500 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	3/4"	150	160°	150	1280	3640	ASME B16.5	INLET
N2	4"	150	115°	150	1280	3640	ASME B16.5	INLET
N3	2"	150	135°	150	1280	3640	ASME B16.5	INLET
N21	6"	150	180°	200	1940	400	ASME B16.5	OUTLET
N22	6"	150	0°	-	-	3640	ASME B16.5	VENT
N10	4"	150	30°	150	1750	70	ASME B16.5	DRAIN
N11	6"	150	30°	200	1750	3100	ASME B16.5	EMERGENCY OVERFLOW
N12	24"	150	90°	300	1280	3640	API 650	INSPECTION HOLE
M1	24"	150	90°	300	1750	850	API 650	MANHOLE
K1	4"	150	45°	300	1280	3640	ASME B16.5	LEVEL INDICATOR
N4	4"	150	180°	150	1280	3640	ASME B16.5	SPARE

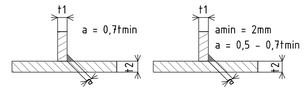
NOTE:
1. REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGHT DESIGN
2. REPORTED SEISMIC LOAD IS A PRIMARY LOAD, DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301 AND IS SUITABLE FOR COMBINATION IN ACCORDANCE WITH SBC301 FOR EITHER STRENGHT DESIGN OR ALLOWABLE STRESS DESIGN. COMBINATION FACTOR FOR STRENGHT DESIGN SHALL BE 1.0, HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN METHOD THEN THE COMBINATION FACTOR WOULD BE 0.7.
3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	01-07-20	FIRST ISSUE	KL	JS	SC	RL	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_M_G_02918_044 — REVISION: 0

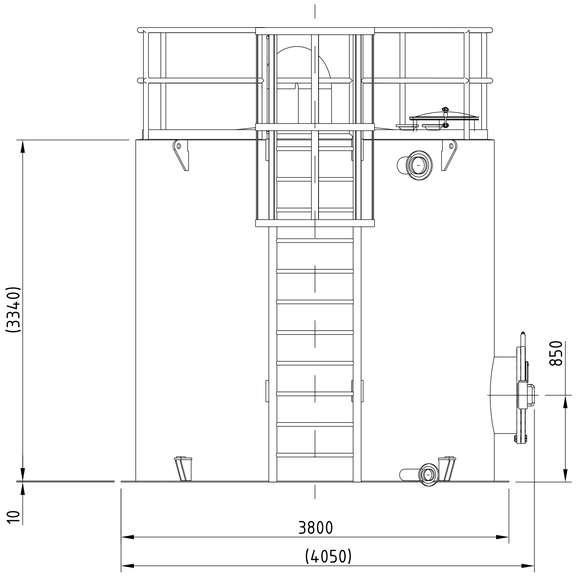
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DRAWN	KimLop	01-07-20
CHECKED	JonSal	01-07-20
ENGINEER	SouCha	01-07-20
APPROVED	RicLeh	01-07-20
EMPLOYER		
SCALE	1:30	A1

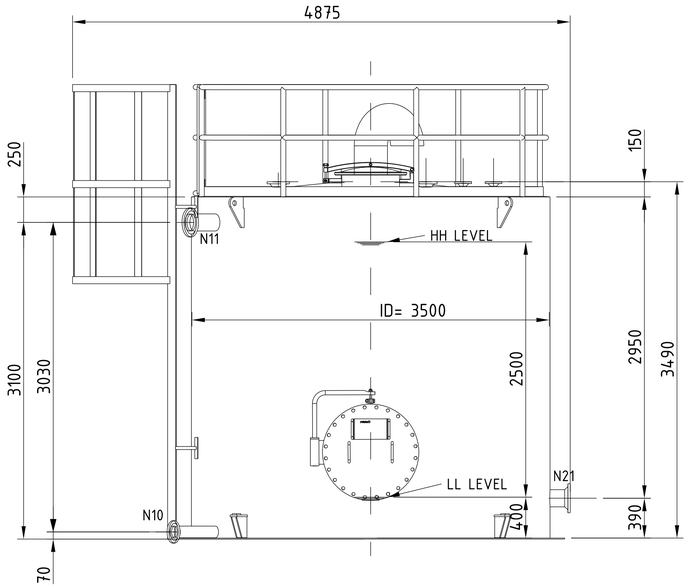
MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
LOW GRADE ACID WASH, ELUTION AND CARBON REGENERATION
3660-TNK-001 LG ACID WASH TANK

DRAWING NUMBER: MD-216-3660-EG-ME-GAD-1034 — REV: A01

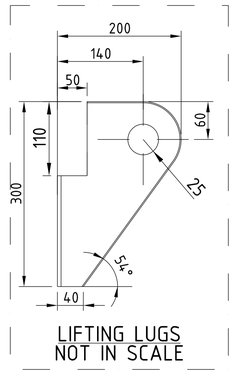
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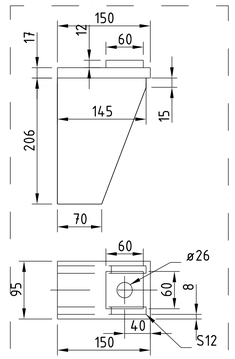
FRONT VIEW



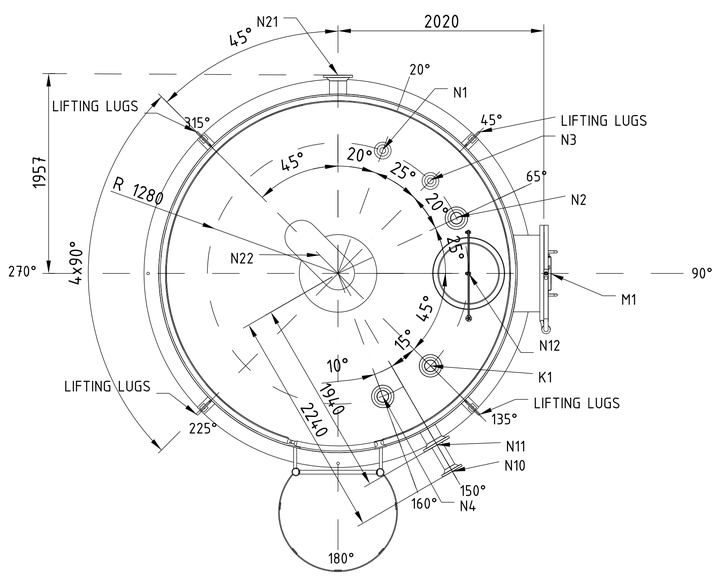
RIGHT VIEW



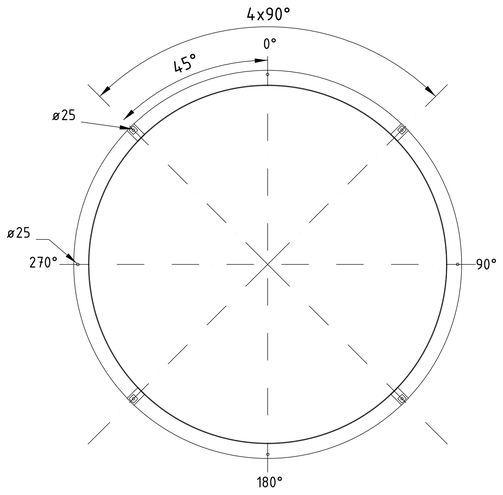
LIFTING LUGS
NOT IN SCALE



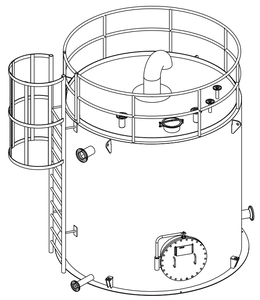
ANCHORING SADDLE
NOT IN SCALE



TOP VIEW



ACHORING



ISOMETRIC VIEW

DESIGN DATA

EQUIPMENT TAG NOs. / NAME PLATE	3660-TNK-004
DESCRIPTION	LG NEUTRALISATION WASH TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	6 mm
BOTTOM THICKNESS	10 mm
CORROSION ALLOWANCE: SHELL INSIDE	2 mm (6 THK. RUBBER LINING PROVIDED)
CORROSION ALLOWANCE: BOTTOM	2 mm (6 THK. RUBBER LINING PROVIDED)
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	4970mmx4050mmx4450mm
SHELL HEIGHT	3350mm
CAPACITY: TOTAL VOLUME	31m³
CAPACITY: LIVE VOLUME	25m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

TOTAL WEIGHT (FULL)	36940 kg
TOTAL WEIGHT (EMPTY)	3840 kg
RUBBER LINING WEIGHT	340 kg
TANK WEIGHT	3500 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO RF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N1	1"	150	20°	150	1280	3490	ASME B16.5	INLET
N2	1 1/2"	150	65°	150	1280	3490	ASME B16.5	INLET
N3	2"	150	45°	150	1280	3490	ASME B16.5	INLET
N21	6"	150	0°	200	1750	400	ASME B16.5	OUTLET
N22	6"	150	0°	-	-	3490	ASME B16.5	VENT
N10	4"	150	150°	485	1750	70	ASME B16.5	DRAIN
N11	6"	150	150°	200	1750	3100	ASME B16.5	EMERGENCY OVERFLOW
N12	24"	150	90°	300	1280	3490	API 650	INSPECTION HOLE
M1	24"	150	90°	300	1280	850	API 650	MANHOLE
K1	4"	150	135°	150	1280	3490	ASME B16.5	LEVEL INDICATOR
N4	4"	150	160°	150	1280	3490	ASME B16.5	SPARE

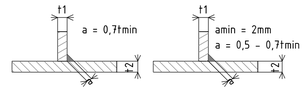
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3. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE STATED.

WELDING GENERAL NOTES

ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING : GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	01-07-20	ISSUED FOR APPROVAL	KL	JS	SC	RL	

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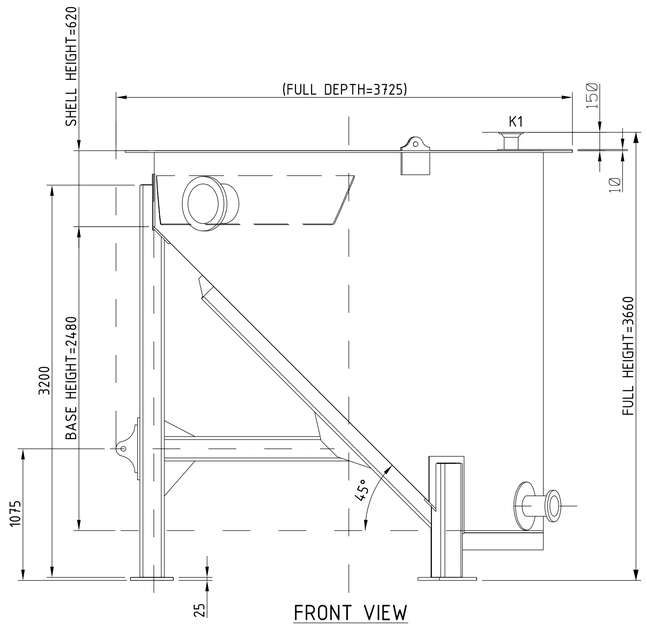
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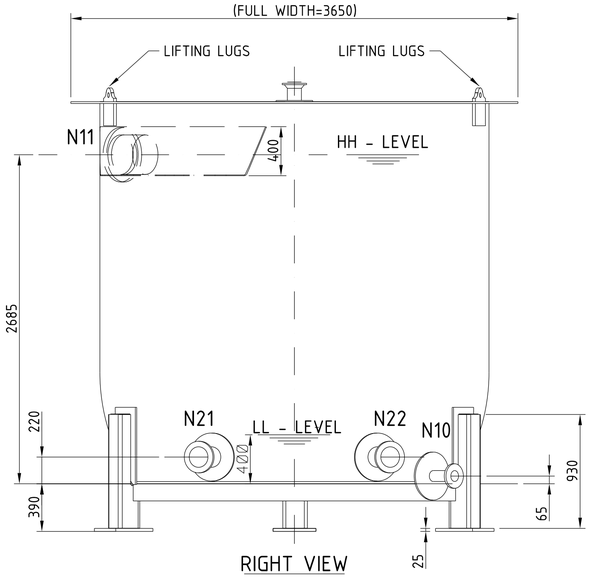
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O19045_450_M_G_02918_047			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707_GAU01_EE13_MTA01_36609		OU100680643	0
DRAWN	KimLop	01-07-20	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	JonSal	01-07-20	OUTLINE DRAWING
ENGINEER	SouCha	01-07-20	LOW GRADE ACID WASH, ELUTION AND CARBON REGENERATION
APPROVED	RicLeh	01-07-20	3660-TNK-004 LG NEUTRALISATION WASH TANK
EMPLOYER			
SCALE 1:30		A1	DRAWING NUMBER MD-216-3660-EG-ME-GAD-1037 REV A01

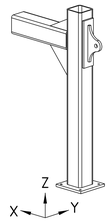
DRAWING NUMBER: MD-216-3660-EG-ME-GAD-1038
REV: A01



FRONT VIEW



RIGHT VIEW



FOOTING COORDINATES

DESIGN DATA

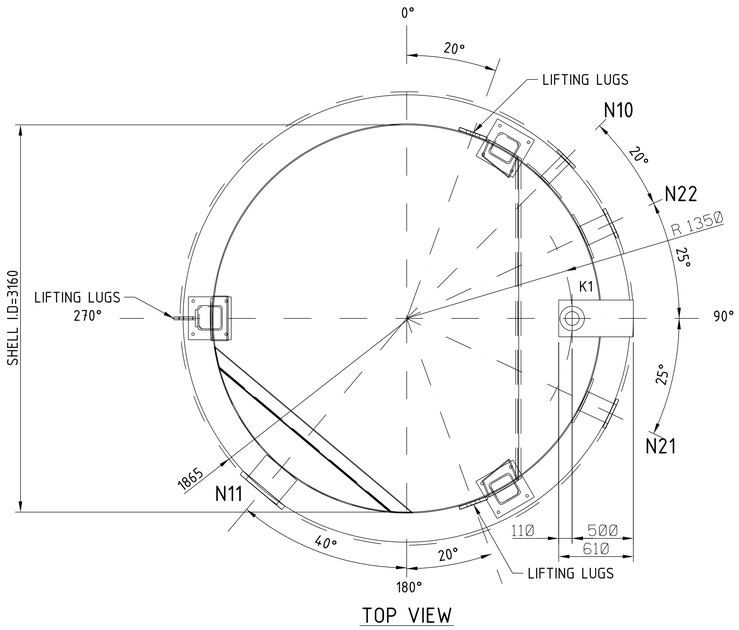
EQUIPMENT TAG NOs. / NAME PLATE	3660-TNK-005
DESCRIPTION	LG PROCESS WATER TRANSFER TANK
DESIGN CODE	API 650
DESIGN TEMPERATURE	+0-60°C
DESIGN PRESSURE	90.97 kPa Absolute
OPERATING PRESSURE	ATM
WIND PRESSURE/ CODE	Site C / 43 m/s / SBC 301,2007
	Kz=ELEV, Kzt=1.0, Kd=0.95
SHELL THICKNESS	8 mm (SEE FEM ANALYSIS)
BOTTOM THICKNESS	18 mm (SEE FEM ANALYSIS)
CORROSION ALLOWANCE: SHELL INSIDE	2 mm
CORROSION ALLOWANCE: BOTTOM	2 mm
TESTING	AS PER APPROVED Q.A.P
RADIOGRAPHY	AS PER APPROVED Q.A.P
JOINT EFFICIENCY	AS PER APPROVED Q.A.P
LOCATION	OUTDOOR
MAIN DIMENSIONS	3650x3725mmx3660mm
CAPACITY: FULL VOLUME	16.5m³
CAPACITY: LIVE VOLUME	10m³
SEISMIC DESIGN /CODE	Site B / SBC 301, 2007
	Ss=0.1g / S1=0.02g / Fa=1.0 / Fv=1.0
	Sds=0.067g / Sd1=0.013g / TL=4s

LOAD DATA

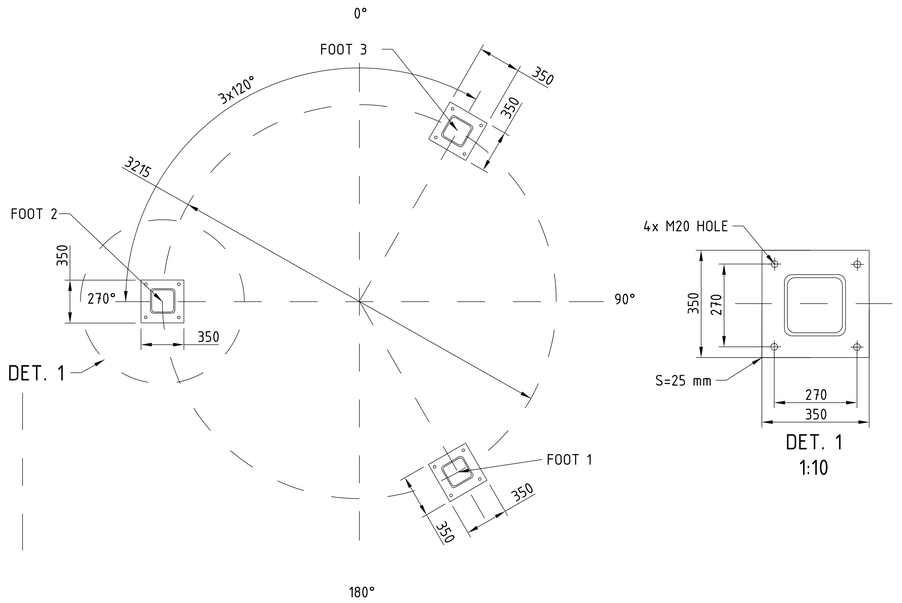
TOTAL WEIGHT (FULL)	23750 kg
TOTAL WEIGHT (EMPTY)	4250 kg
RUBBER LINING WEIGHT	NO LINING
TANK WEIGHT	4250 kg
DETAIL WEIGHTS AND MATERIAL THICKNESSES ACCORDING TO VENDOR DETAIL ENGINEERING	

MATERIAL OF CONSTRUCTION

PLATE - BOTTOM/SHELL/LIFTING LUGS	ASTM A284/ EN10025-2 S235JR
STRUCTURALS	ASTM A284/ EN10025-2 S235JR
NOZZLE	ASTM A284 / A106 Gr B / A53 Gr B
NOZZLE FLANGES	ASME B16.5 SO FF, ASTM A105
SURFACE TREATMENT	EN ISO 12944-5: EP240/4 - FE SA 2½
RUBBER LINING (INNER SURFACES)	NATURAL RUBBER LINER: HARDNESS ±40 ShA



TOP VIEW



ANCHOR PLATES

DET. 1
1:10

NOZZLES TABLE

POS	SIZE	RATING	DIRECTION	PROJECTION	OFFSET	ELEVATION	STANDARD	SERVICE
N21	6"	150	115°	270	1580	220	ASME B16.5	OUTLET
N22	6"	150	65°	270	1580	220	ASME B16.5	OUTLET
N10	4"	150	45°	270	1580	65	ASME B16.5	DRAIN
N11	8"	150	220°	270	1580	2685	ASME B16.5	EMERGENCY OVERFLOW
K1	4"	150	90°	160	1350	3200	ASME B16.5	LEVEL INDICATOR

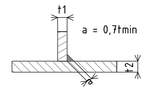
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WELDING GENERAL NOTES

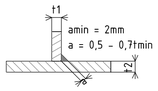
ALL WELDING SEAMS , MUST BE WELDED CONTINOUSLY UNLESS OTHERWISE SPECIFIED.

TYPICAL FILLET WELD THICKNESS:

ONE SIDE WELDING :



GENERAL :



EXECUTE WELDING WORK ACCORDING TO DIN EN ISO 5817:
- FOR BUTT WELDS, QUALITY LEVEL C,
- FOR FILLET WELDS, QUALITY LEVEL B,

GENERAL TOLERANCE FOR WELDED CONSTRUTION ACCORDING TO EN ISO 13920 - AE.

GENERAL TOLERANCE acc. to ISO 2768 mH

MM	Machined MM	none Machined MM	Straightness Flatness MM
<= 6	± 0.1	± 0.5	0.008
> 6 - 30	± 0.2	± 1.0	0.020
> 30 - 120	± 0.3	± 1.5	0.040
> 120 - 400	± 0.5	± 2.0	0.080
> 400 - 1000	± 0.8	± 3.0	0.150
> 1000 - 2000	± 1.2	± 4.0	0.200
> 2000 - 4000	± 2.0	± 6.0	
> 4000	± 3.0	± 8.0	

CONCENTRIC and AXIAL RUN TOLERANCE 0.1 MM

SYMMETRIC TOLERANCE 0.5 MM

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3310-EG-ME-RPT-1002	3310-TNK-007 FEM ANALYSIS REPORT	A01	01-07-20	FIRST ISSUE	KL	JS	SC	RL	

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OUTOTEC DRAWING NUMBER: 900707_GAU01_EE13_MTA01_36610 — DOCUMENT ID: 0U100680644 — REVISION: 0

DRAWN	KimLop	01-07-20
CHECKED	JonSal	01-07-20
ENGINEER	SouCha	01-07-20
APPROVED	RicLeh	01-07-20
EMPLOYER		
SCALE	1:25	A1

MANSOURAH - MASSARAH GOLD PROJECT
OUTLINE DRAWING
LOW GRADE ACID WASH, ELUTION AND CARBON REGENERATION
3660-TNK-005 LG PROCESS WATER TRANSFER TANK

DRAWING NUMBER: MD-216-3660-EG-ME-GAD-1038 — REV: A01

ATTACHMENT 2 VENDOR CALCULATION FOR 3660-TAN-005 LOADING

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O19045 - MMGP - 3660 FDN LOADING REQUIREMENT

Inbox ×

Sonjit Mallick
Dear Sourav The clarification as marked on attached drawing is urgently required, else we cannot proceed design. Request for

Sonjit Mallick
to me

From: Joni Salo <joni.salo@mogroup.com>
Sent: 02 November 2020 19:28
To: Sonjit Mallick <sonjit@lntecc.com>
Cc: Sourav Chattopadhyay <sourav.chattopadhyay@mogroup.com>; Tatu Vierma <tatu.vierma@mogroup.com>; Luis Pizarro <lui
Subject: RE: O19045 - MMGP - 3660 FDN LOADING REQUIREMENT

CAUTION: This is an external email. Do not click links or open attachments unless you recognize the sender and know the

Hello Sonjit,

The outline drawing is referencing the FEM report which has all the footing loads. Attached, pages 35-37.

MD-216-3310-EG-ME-RPT-1002 3310-TNK-007 FEM ANALYSIS REPORT

- Joni

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Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **FEM ANALYSIS REPORT**

GOLD PROCESSING PLANT
FLOTATION
3310-TNK-007 - CLEANER TAILS PUMP SUMP 15m3

Document Number: **MD-216-3310-EG-ME-RPT-1002**
Revision: **A01**
Revision Date: **28.11.2019**
Status: **ISSUED FOR INFORMATION**

Outotec Document Number:	900707	GAU01	ZF01	MED01	01002	OU100667622	0
L&T Document Number:							

REVISION HISTORY						
Rev	**Date**	**Revision Text**	**Prepared By**	**Reviewed By**	**Approved By**	**Employer**
A01	28.11.2019	FIRST ISSUE	tuopul	SouCha	t_ricleh	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
'EMPLOYER'	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS, American Bechtel Inc. or their representatives
'Contractor '	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'MMGP Project'	:	Mansourah Massarah Gold Project
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
Shall	:	Is to be understood as mandatory in relation to the requirements of this Document
Should	:	Is to be understood as a strong recommendation to comply with the requirements of this Document
Quality Assurance (QA)	:	Part of quality management focused on providing confidence that quality requirements will be fulfilled (ISO 9000:2015)
Quality Control (QC)	:	Part of quality management focused on fulfilling quality requirements (ISO 9000:2015)
Area	:	Physical or logical division of the process unit according to the discipline and as defined by EPC.
CE	:	Conformité Européene/ European Conformity
ISO	:	International Standards Organization
FEA	:	Finite Element Analysis

1. INTENT

The intent of this technical report is to assess the static and buckling strength of the 3310-TNK-007 - Cleaner Tails Pump Sump (15m3) to be supplied for **Ma'aden Mansourah Massarah Gold Project** in the Kingdom of Saudi Arabia.

This technical report also presents the loads which arise from operational and environmental conditions and are applied to foundation. Lifting analysis was done and required lifting lug dimensions were determined.

FEMAP 12.0 with NX Nastran was used for pre/post processing and analysis and determine the static and buckling strength of 3310-TNK-007 - Cleaner Tails Pump Sump.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week. Cleaner Tails Pump Sump was analysed for required shell thicknesses and foundation loads.

Shell thickness calculations were made in accordance to API 650. Environmental loads were acquired from Ma'aden Civil, Structural and Architectural Design Criteria (*MD-216-1000-EG-ST-CRT-1001)* and approved Site Condition document (*MD-216-1000-EG-GE-SPC-1001*). Cleaner Tails Pump Sump is classified as a process building and analysed accordingly.

Environmental loads are according to SBC 301 (2007) and safety factor of 2.5 was used for lifting lugs. Material used in Cleaner Tails Pump Sump is EN 10025-2 S235JR, which is common carbon alloy steel used widely in steel structures.

Cleaner Tails Pump Sump meets the design requirements and can be operated safely.

3. SCOPE

3.1. General

The static and buckling strength of the tank and support structure were analysed. The strength of the lifting lugs was also checked.

Weight of the structure and its contents were used as the loading in static strength assessment. Also, wind load was included. Additional nozzle load was applied for the nozzle at the bottom of the tank shell.

According to appendix E of code API 650: “Tanks located in regions where S1 is less than or equal to 0.04 and SS less than or equal 0.15, or the peak ground acceleration for the ground motion defined by the regulatory requirements is less than or equal to 0.05g, need not be designed for seismic forces; however, in these regions, tanks in SUGIII shall comply with the freeboard requirements of this appendix.” The seismic horizontal acceleration (according to SBC 301) is 0.027g (C_s). Therefore, seismic strength study is not required. However, the horizontal acceleration together with structural mass and sanded liquid mass will be used to determine the support reactions to footing.

3.2. Code and Specifications

1. API 650-10 Welded tanks for oil storage, 2010.
2. Wind load based on SBC 301 – Structural - loading and forces.
3. Seismic load determination based on SBC 301.

3.3. Design criteria

1. Structural Design Criteria – Process Buildings and Structures: MD-101-SMEM-EG-ST-CRT-0001
2. Civil, Structural and Architectural Design Criteria: MD-216-1000-EG-ST-CRT-1001 (900707_GAU01_ZZ01_CEC03_00001_OU100656130)
3. GEOTECHNICAL INVESTIGATION REPORT: MD-216-49RA-EG-CV-RPT-1001 (900707_GAU01_AN03_CQS03_49001_OU100664631)

3.4. Drawings

1. OUTLINE DRAWING - 3310-TNK-007: CLEANER TAILS PUMP SUMP (15M3) : MD-216-3310-EG-ME-GAD-1010 (*900707_GAU07_ZF01_MTA01_33106_OU100660892*)

3.5. Design criterion, allowable product design stress and interpretation of FEA results

- According to API STANDARD 650 – “Welded Tanks for Oil Storage”, the maximum allowable product design stress for grade 235 steel is 137 MPa (= two-thirds the yield stress or two-fifths the tensile strength, whichever is less).
- The given allowable product design stress is intended to be used in analytical calculations of the tanks.
- The analytical calculations consider only the circumferential primary membrane stress of the shell and do not take into account bending stresses, secondary stresses or local stress concentrations.
- However, when utilizing FEA, the obtained stresses are including all these before mentioned stresses.
- In this report, while interpreting the stress results, following aspects are taken into account:
    - Local stress concentrations are ignored if the neighbouring areas have sufficient capacity to carry the external load without significant plastic deformations.
    - The allowable product design stress for the bending type of stress is 1.5 times the allowable product design stress for the membrane type of stress.
    - Secondary stresses (e.g. due to enforced motion) are ignored, and primary stresses are used for judging the eligibility of the structure.
    - In order to avoid the risk of cyclic plasticity, exceeding three times the product design stress is not allowed.

4. FE-MODEL

4.1. Geometry

Geometry file on the basis of analyses is *3310-TNK-007 CLEANER TAILS PUMP SUMP 15m3*. Some changes were performed for the structure (The dimensions in modifications of the figures are reported from the mid surfaces of the plates):

1. The extension of the shell at the front bottom was removed (fig. 4.1.1).
2. The thickness of the tank shell was decreased PL6 (fig. 4.1.2)
3. The thickness of the tank bottom was decreased PL13 (fig. 4.1.2).
4. The lifting lugs of the upper edge of the tank is presented in figure 4.1.3.
5. The thickness of the reinforcement plate below the lower lifting lug was decreased to PL15. figure 4.1.3
6. The width of the stiffening ring (PL15) at the upper edge was decreased to 200mm (originally 245 mm). (See figure 4.1.4)

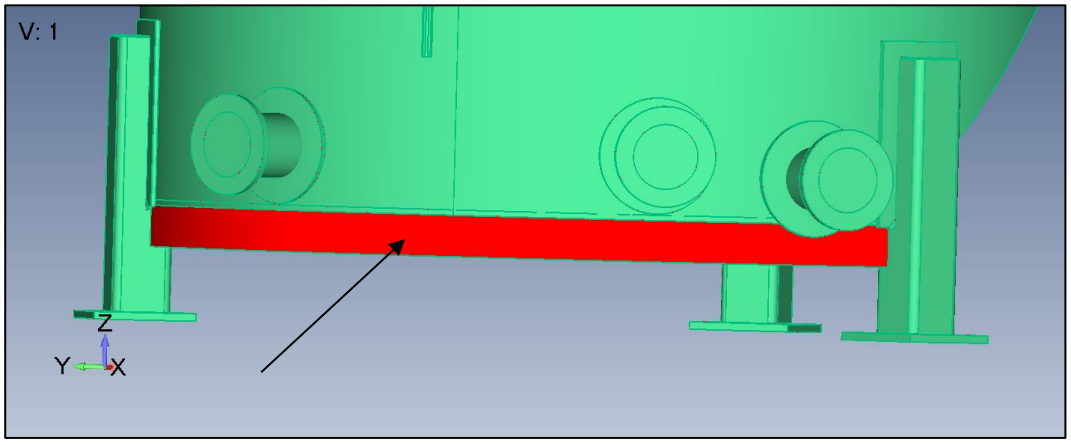


Figure 4.1.1. The extension of the shell (part with red colour) was removed.

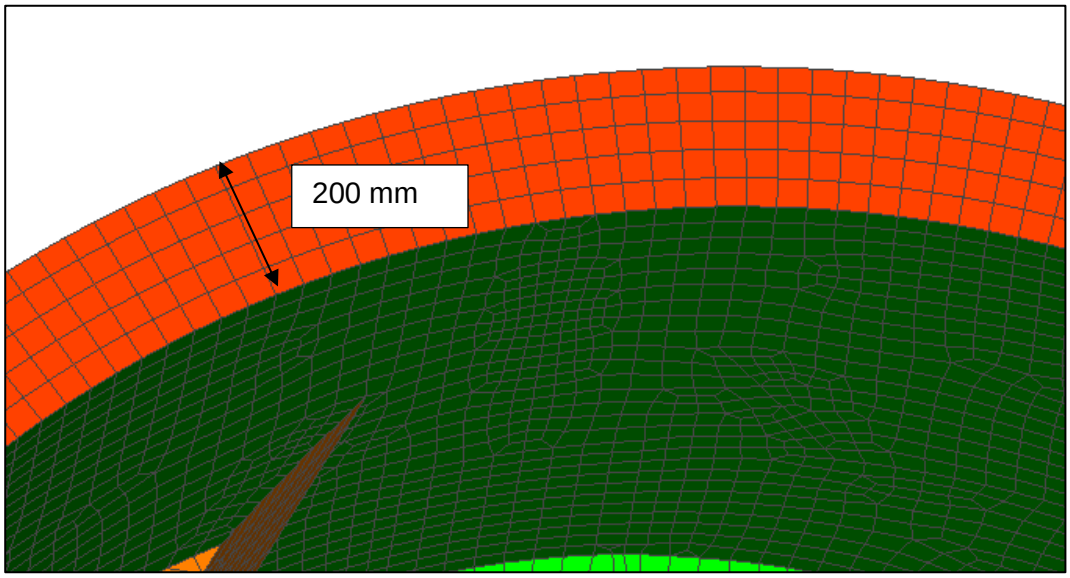


Figure 4.1.4. The width of the stiffening ring (PL15) was decreased to 200 mm.

4.2. FE-model

The whole tank with stiffeners, nozzles, support structure and lifting lugs were analysed.

Analysis types were linear elastic static and buckling analysis.

Parts were mainly (plates and welds) modelled using linear shell elements generated at the mid surfaces of the respective plates. The welds were modelled equal strength to parent material. RBE3 interpolation element was used to attach nozzle load. Lifting lugs at the upper edge were modelled with solid elements.

FE-model is presented in Figures 4.2.1 and 4.2.2. The plate thicknesses are reported with analysis thickness. The corrosion allowance (2 mm) shall be added to the presented thicknesses where needed.

The FE-model consists of 42 100 elements and 42300 nodes.

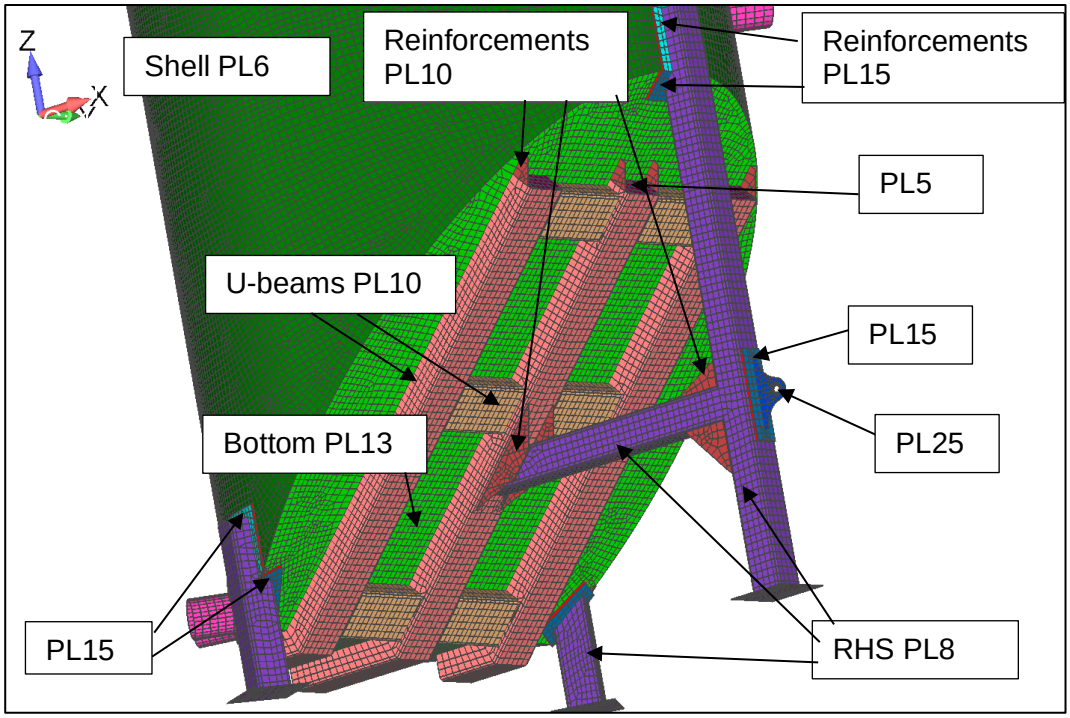


Figure 4.2.1. FE-model of the tank, view from the bottom.

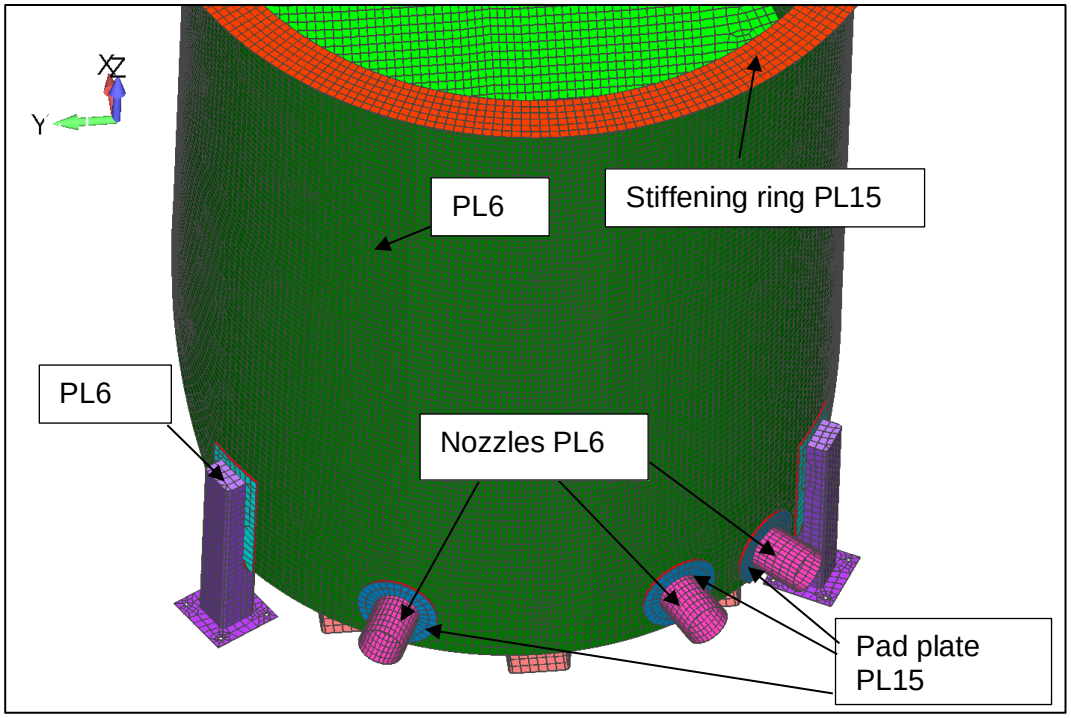


Figure 4.2.2. FE-model of the tank, view from the top.

4.3. Material model and properties

All materials were modelled as linear-elastic. The material of the structure is P235/S235 steel.

Modulus of elasticity for 235 steel is E = 210 GPa, density 7850 kg/m3 and Poisson's ratio v = 0.3.

The minimum yield strength of the material (S235/P235) is 235 MPa, the minimum tensile strength is 365 MPa and the design stress 137 MPa.

4.4. Boundary conditions

The boundary conditions of the structure in other load cases than lifting are presented in figure 4.4.1. The structure was constrained from the legs with fixed constraints. Boundary conditions in lifting are presented in section 4.5.2.

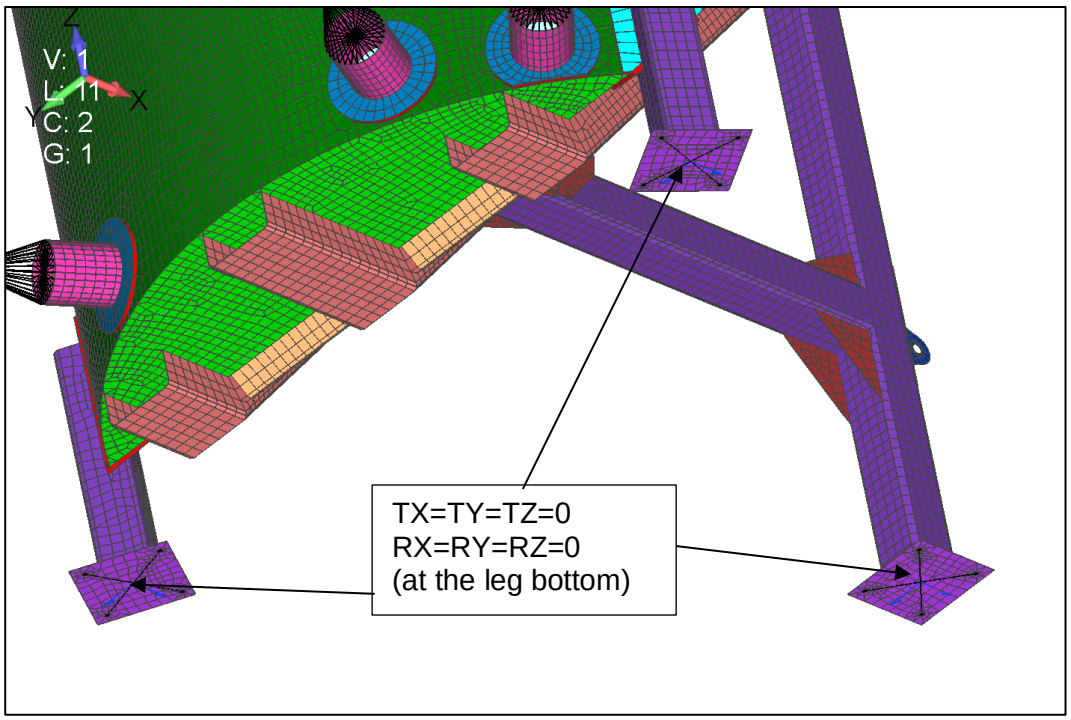


Figure 4.4.1. Boundary conditions of the structure (other load cases than lifting).

4.5. Loading

The loadings of the structure consist of structural weight, weight of the contents, additional nozzle load, wind load and horizontal acceleration (seismic load).

4.5.1. Loads

Self-weight of the structure was applied to the structure as material density and global acceleration (g = 9.81 m/s^2). Self-weight was multiplied by the factor 1.1 to compensate the missing masses. Also, mass of the corrosion allowance (2 mm, 0.67 ton) was compensated with non-structural mass. Therefore, the total structural mass of the tank was 4.9 ton.

The content is sanded liquid with density ρ = 1.44 t/m^3 was applied. The liquid level is presented in figure 4.5.1.

Self-weight of the contents was applied as hydrostatic pressure including reaction forces at the nozzles.

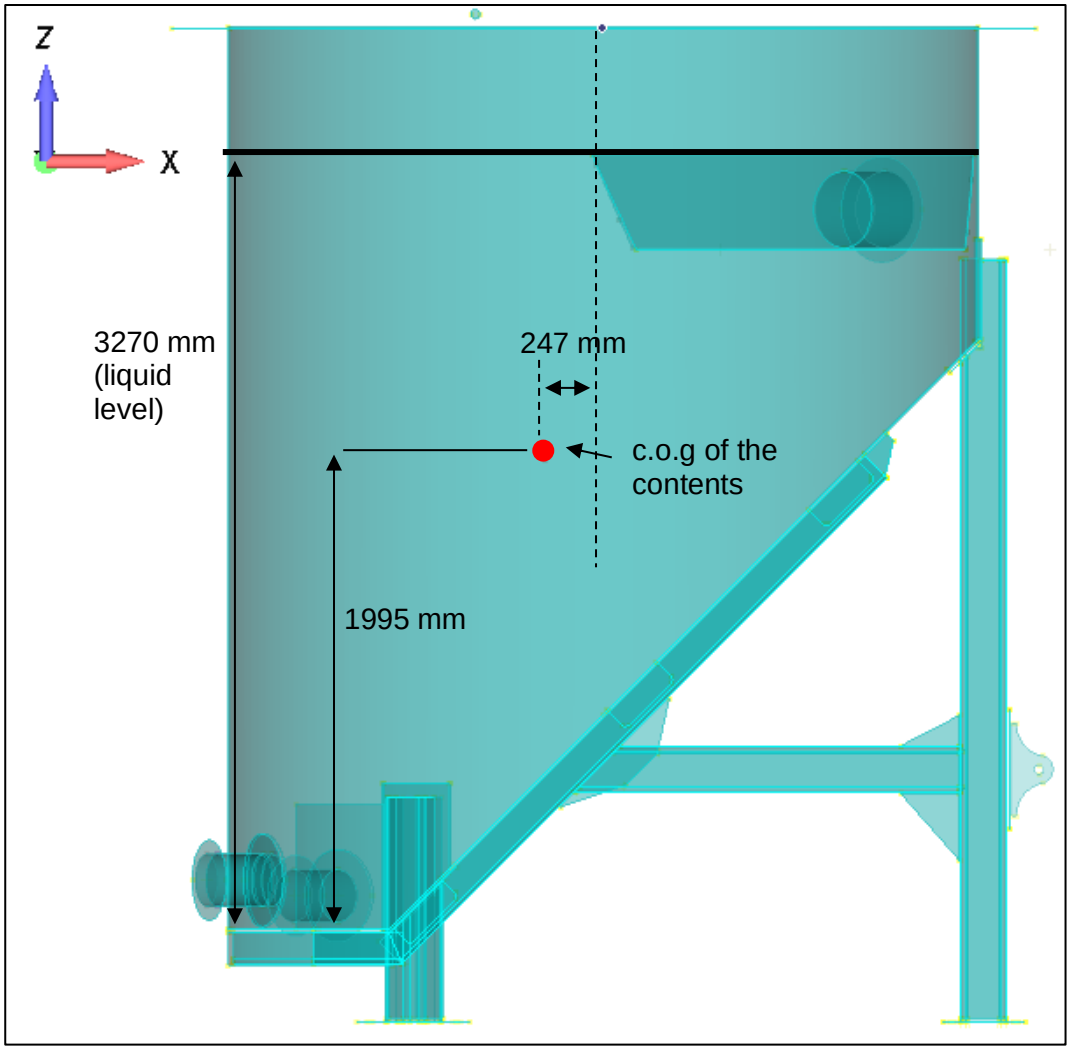


Figure 4.5.1. Liquid level at the normal operational conditions.

Additional nozzle load (figure 4.5.2) was applied for the nozzles at the bottom part of the tank shell. The force was attached to the structure with RBE3 interpolation element.

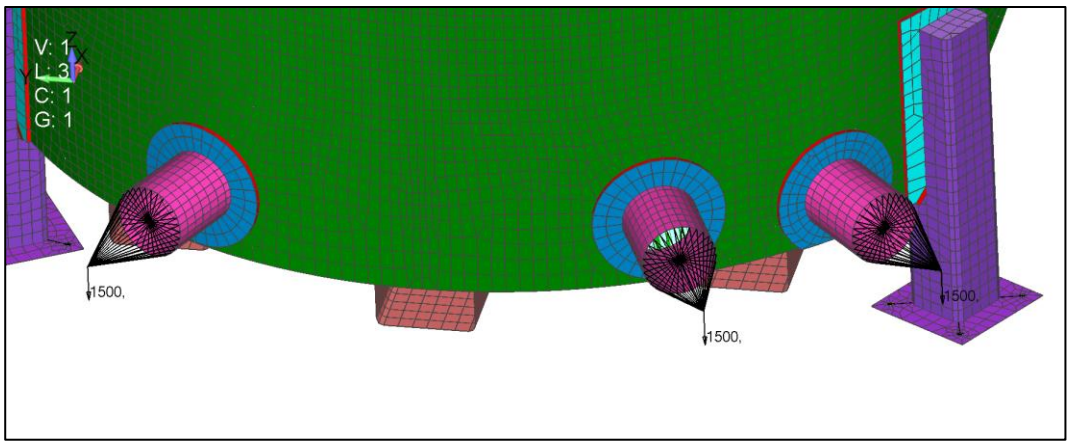


Figure 4.5.2. Additional nozzle load (1.5 kN downwards at the distance of 300 mm from the flange) for the nozzles at the bottom part of the tank shell.

Wind load was determined according to SBC 301 section 7.2.

Occupancy classification of tank is made to category II (Table 1.6-1, SBC 301). It is assumed that release of hazardous material does not pose a threat to the public.

The wind directionality factor, $K_d = 1.0$.

Exposure category C. Velocity pressure exposure coefficient

$K_z = (K_h) = 0.86$ (height above ground level z = 5 m)

Importance factor I = 1.00 (Table 6.5-1, SBC 301).

It is assumed that the topographic factor is $K_{zt} = 1.00$. (flat ground)

With the wind load the tank is taken as simple cylinder and is classified as closed building.

Building is assumed to be rigid (fundamental frequency is greater than or equal to 1 Hz). Therefore, gust effect factor is taken as G = 0.85.

Basic wind speed V = 155 km/h (43 m/s).

Velocity pressure at height (z = 5 m)

$$q_z = 0.0473*10^{-3} * Kz*Kzt*Kd*V^2*I \text{ (kN/m}^2\text{)} = 0.977 \text{ kN/m}^2.$$

External pressure coefficient as force coefficient C_f (Figure 7.2-15, SBC 301):

$$C_f = 0.81 \text{ (h/D = 1.3, D*}\sqrt{q_z}\text{=3.98*}\sqrt{0.977}\text{=3.94, D'/D = 0.08)}$$

Based on above parameters design wind force can be defined:

$$F = q_z * G * C_f * A_f = 0.977 \text{ kN/m}^2 * 0.85 * 0.81 * 12.8 \text{ m}^2 = 8.6 \text{ kN}$$

where

A_f = 4.0 m * 3.2 m = 12.8 m^2, projected area normal to the wind (the largest projected area).

Internal pressure coefficient GC_{pi} = -0.18 and internal pressure $p_i = q_z * GC_{pi}$ = - 0.18* 0.977 kN/m^2 = - 0.18 kN/m^2. Is applied to tank structure simultaneously with design wind force F (only in buckling strength assessment).

Seismic load (only for footing loads) was determined according to SBC 301.

The maximum considered earthquake, 5% damped, spectral response acceleration at short periods adjusted for site class effects as defined in Section 9.4.3.

$S_{MS} = F_a S_s$,

S_s = 0.10g

F_a = 1.0 (Table 9.4.3a SBC 301 and site class B, Geotechnical report)

The maximum considered earthquake, 5% damped, spectral response acceleration at a period of 1 sec adjusted for site class effects as defined in Section 9.4.3.

$S_{M1} = F_v S_1$.

S_1 = 0.02g

F_v = 1.0 (Table 9.4.3b SBC 301 and site class B, Geotechnical report)

The design spectral response accelerations (spectrum presented in fig 4.5.3)

S_{DS} = 2/3 S_{MS} = 2/3 * 1.0 * 0.1g = 0.067g

S_{D1} = 2/3 S_{M1} = 2/3 * 1.0 * 0.02g = 0.013g

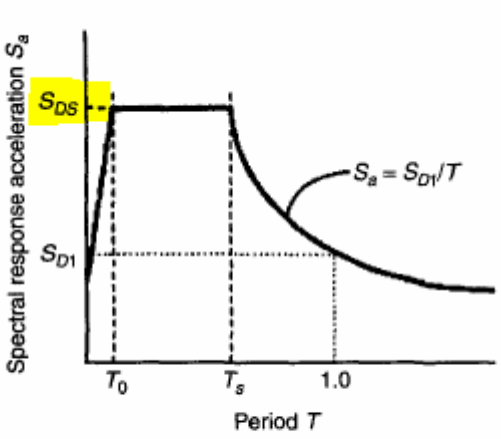


FIGURE 9.4.5
DESIGN RESPONSE SPECTRUM

Figure 4.5.3. Design response spectrum (SBC 301).

For seismic design category B equivalent lateral force according to SBC 301 section 10.9 was determined.

Seismic base shear is determined according to equation

$$V = C_s * W$$

where W = total operational weight (structural + contents)

$C_s = S_{DS}/(R/I) = 0.027g$ (maximum acceleration S_{DS})

R = 2.5 (SBC 301, Table 10.2 (structural steel system not specifically detailed for seismic resistance)

I = 1.0 (Occupancy category II, SBC 301 Table 1.6-1)

When calculating the support reactions for footing, the horizontal force of the contents (V = 0.027*g*26.4 ton = 7.0 kN) was applied to the centre of gravity of the contents and attached to the tank shell with RBE3 interpolation element. The horizontal load from the structural weight of the structure was applied as horizontal acceleration of C_s (0.027*g*4.90 ton = 1.3 kN). Total horizontal (seismic) force is therefore V_{tot} = 8.3 kN.

4.5.2. Load combinations

Strength in operational condition will be calculated with load combinations:

LC1 = Structural self-weight + contents + additional nozzle load

LC2 = Structural self-weight(empty) + additional nozzle load

+ wind load (directions in fig 4.5.4)

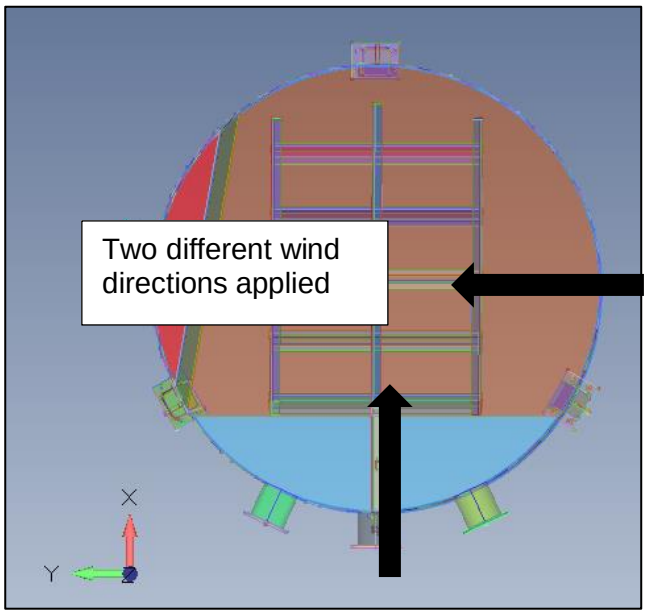


Figure 4.5.4. Different directions for wind load (+x and +y – axis directions).

Lifting situation was analysed with 2.5*Self weight of the structure in two different positions of the structure, see fig 4.5.5. Lifting chain angle is presented in figure 4.5.5. Lifting situation load combination:

LC3 = 2.5*Structural self-weight (empty)

(Note: In the lifting situation the safety factor is 2 (for areas far away from lifting lugs) and 2.5 (for lugs and welds)).

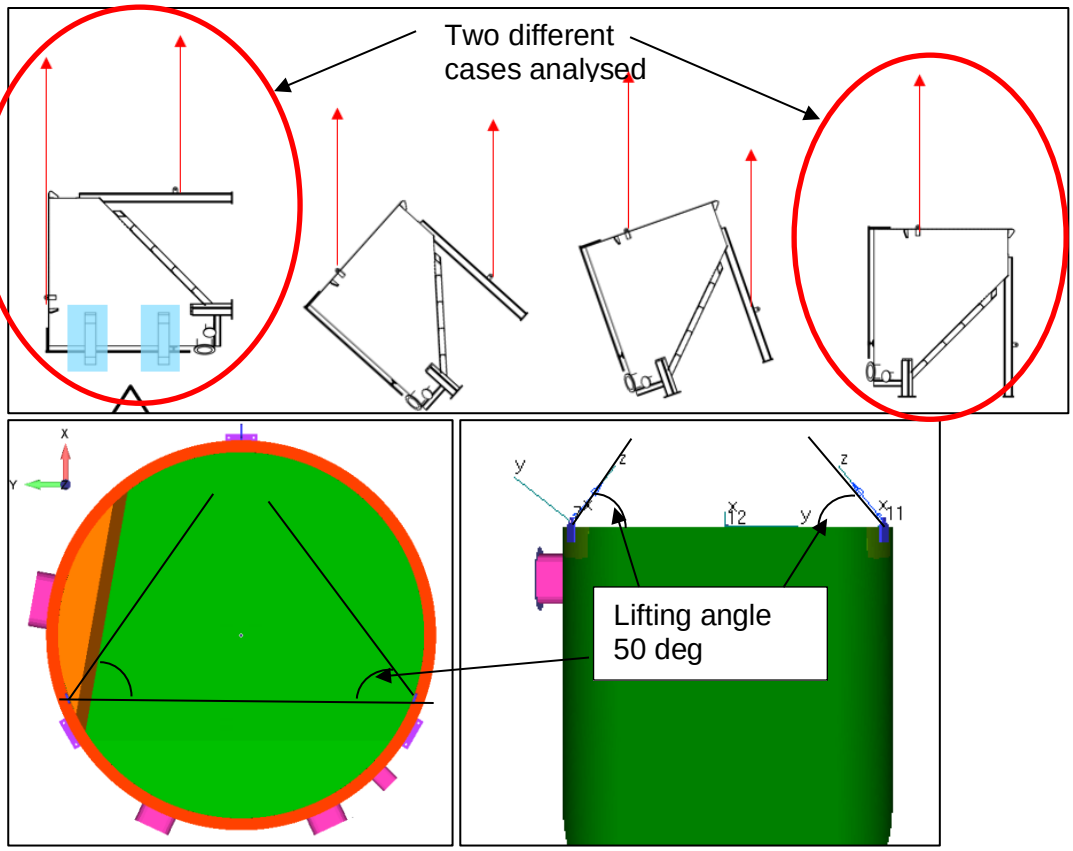


Figure 4.5.5a. Lifting situation. (Global acceleration of gravity was applied towards center(line) of gravity of the structure)

Boundary conditions in lifting situations are presented in figure 4.5.5b. Constraints were applied to the direction of the lifting angle. Also, movement in transverse direction was fixed.

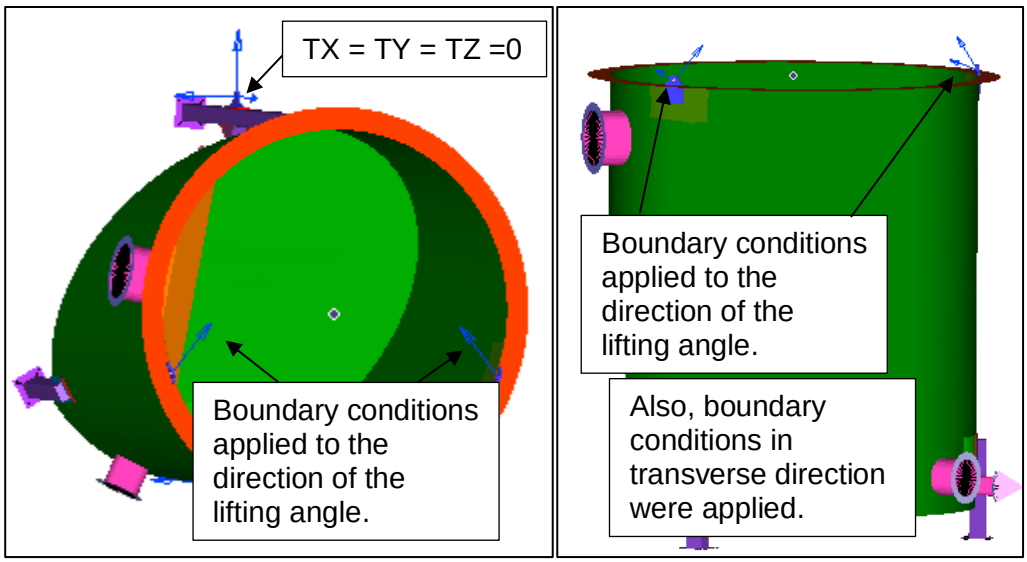


Figure 4.5.5b. Boundary conditions in lifting situations. Left: horizontal position, Right: vertical position.

The support reactions for footing were determined with horizontal (seismic) load: LC4 = Structural self-weight + contents + horizontal (seismic) load (V_{tot}) (four different directions presented in fig 4.5.6)

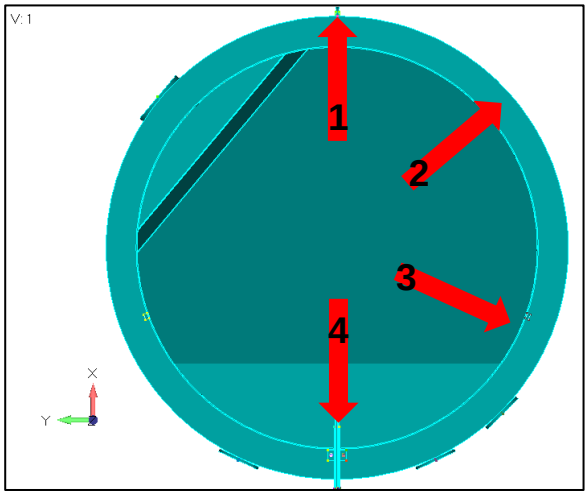


Figure 4.5.6. Four different directions of the seismic load.

Buckling strength for shell was analysed with load combination:

LC5 = Structural self-weight(empty) + wind load (with internal pressure)

In addition, the support reactions for footing were analysed with full of solids condition:

Structural self-weight + contents (100%solids with density 2.76 t/m^3)

5. ANALYSIS RESULTS

5.1. Displacements

Total translation of the structure in the load combination LC1 (operational condition) is presented in figure 5.1.1. Maximum total translation is 2.2 mm at the bottom of the tank.

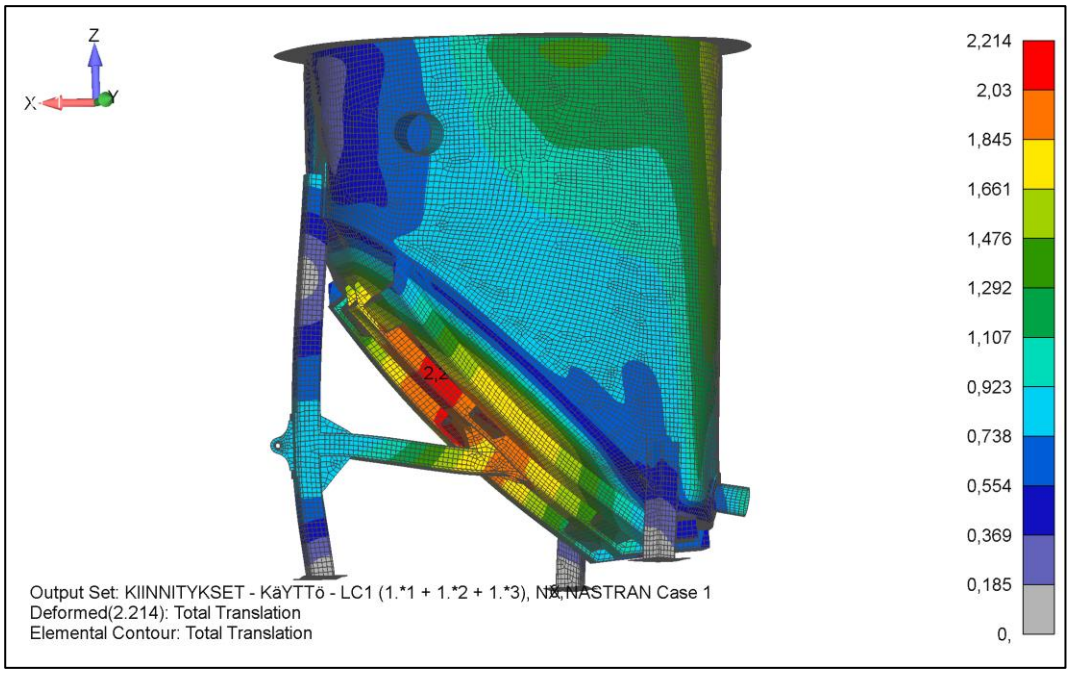


Figure 5.1.1. Displacements of the structure [mm]. Deformation is scaled by the factor of 100.

5.2. Static analysis

The static strength was assessed with two different load combinations LC1 and LC2. The stresses of the structure in the load combination LC1 are presented in figures 5.2.1 – 5.2.6. The von Mises stress exceeded the design stress in several locations maximum von Mises stress being 207 MPa (see fig 5.2.1 and 5.2.4). Utilization factor (UF) based on von Mises stress compared to design stress (137 MPa) is marked to the most critical locations in figures.

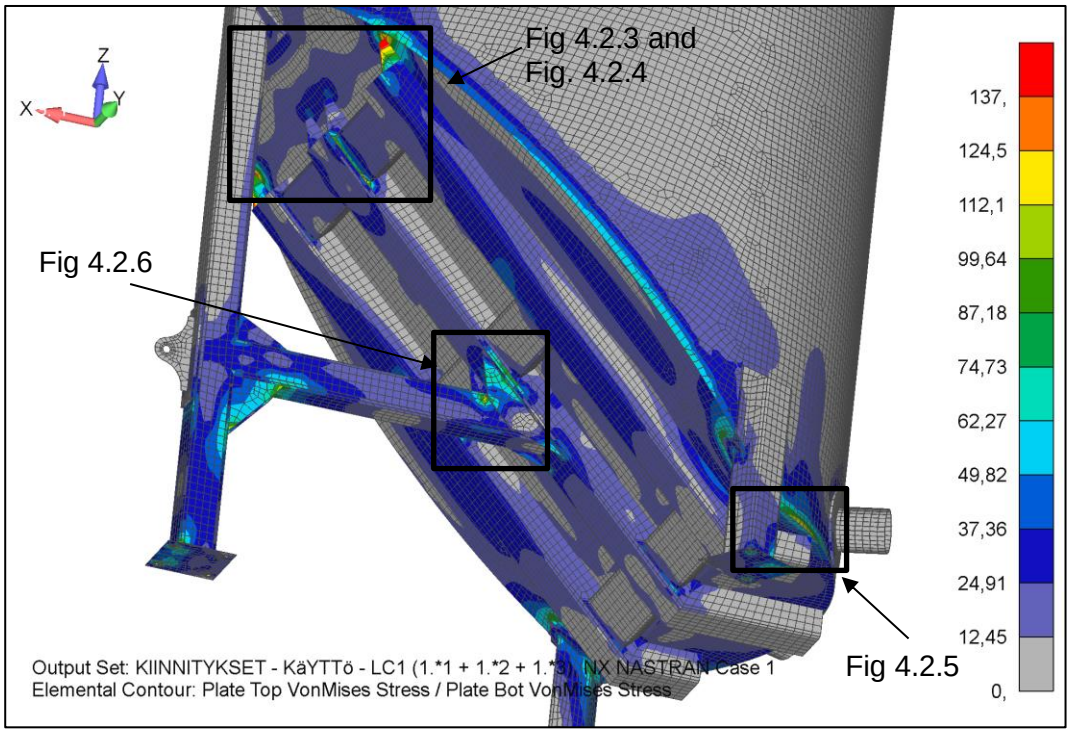


Figure 5.2.1. Von Mises stresses of the structure [MPa] in the operational load combination LC1 (- y-axis direction view).

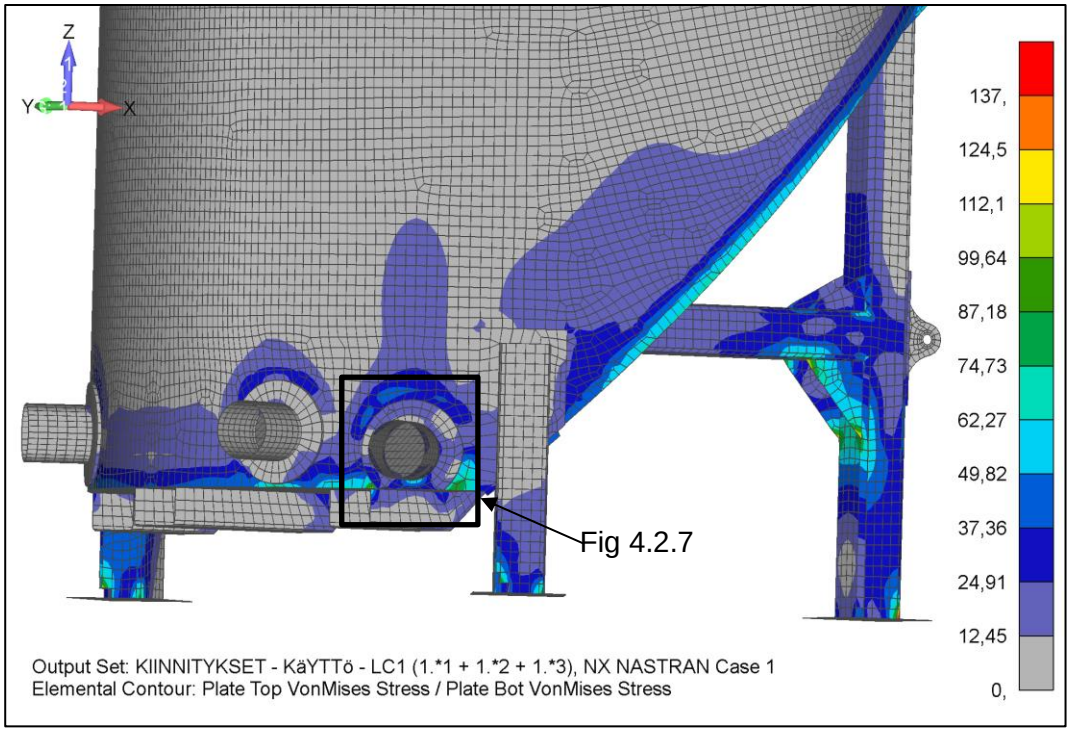


Figure 5.2.2. Von Mises stresses of the structure [MPa] in the operational load combination LC1 (+ y-axis direction view).

The von Mises stress distribution at the upper end of U-beams below the bottom plate is presented in figure 5.2.3. At the edge of the reinforcement plate the maximum stress is 207 MPa. At the bottom of the tank the stress, 154 MPa (at the joint to reinforcement plate), is exceeding the design stress but the exceeding is local. For bending stress 1.5 * 137 MPa = 206 MPa is accepted, and for local stress concentration, 3.0 * 137 MPa = 411 MPa is allowed as long as the highly stressed area is small enough for neighbouring areas to carry the load. The stresses are thus on an acceptable level.

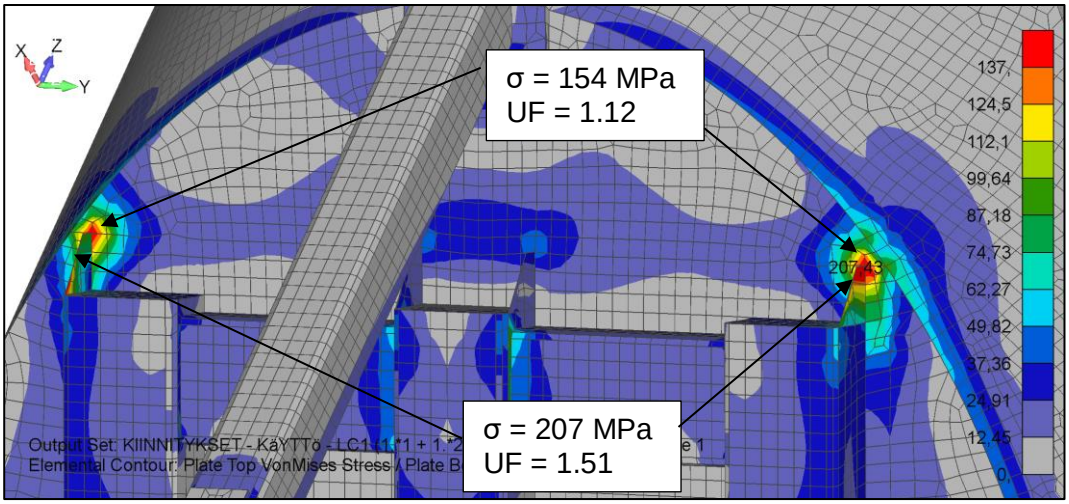


Figure 5.2.3. Von Mises stresses of the structure [MPa] in the operational load combination LC1, at the end of transverse beam.

The von Mises stress distribution at the shell plate (near the upper end of U-beams) is presented in figure 5.2.4. The stress is exceeding the design stress locally with stress peak of 163 MPa. The von Mises stress at the bottom front corner of the shell is presented in figure 5.2.5. The stress is exceeding the design stress locally with stress peak 151 MPa. At both locations the shell is bending around the bottom edge and therefore 206 MPa is accepted.

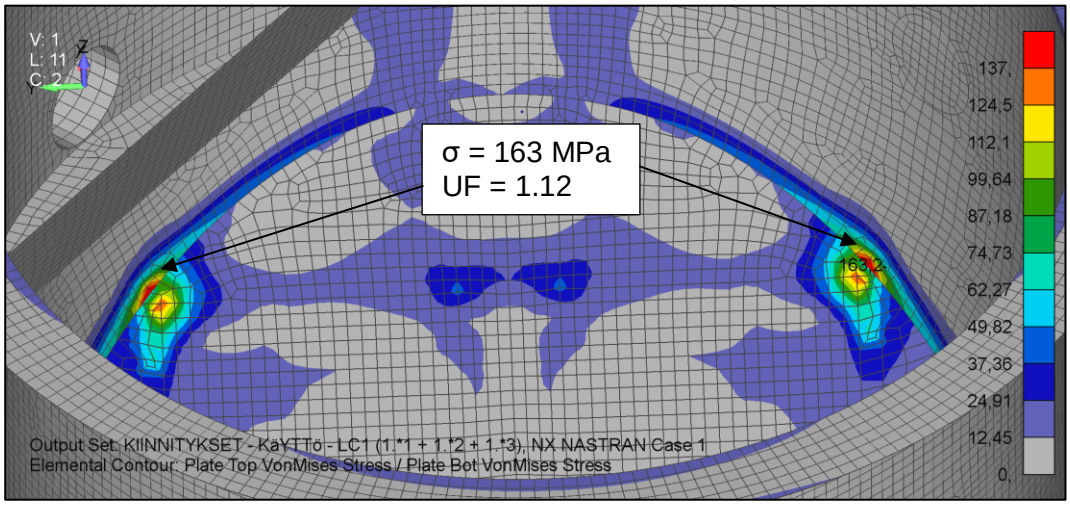


Figure 5.2.4. Von Mises stresses of the structure [MPa] in the operational load combination LC1, inside the shell (near the upper end of the U-beams).

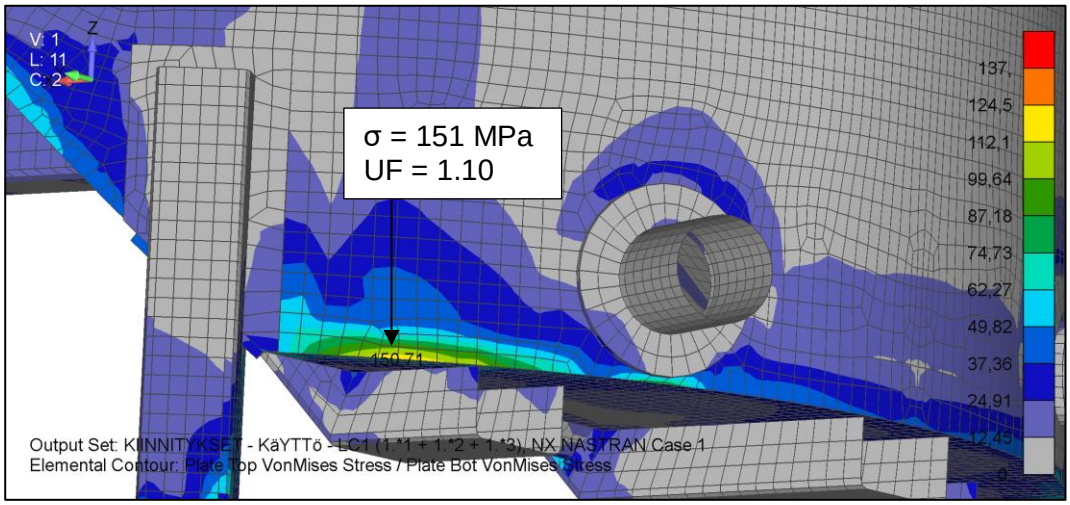


Figure 5.2.5. Von Mises stresses of the structure [MPa] in the operational load combination LC1, at the bottom front corner of the shell.

Stresses at the end of the transverse RHS beam are presented in figure 5.2.6. At the joint of the RHS beam and the U-beam the maximum von Mises stress was 145 MPa at the edge of the reinforcement plate. The stress is local bending of the reinforcement plate and 206 MPa can be allowed.

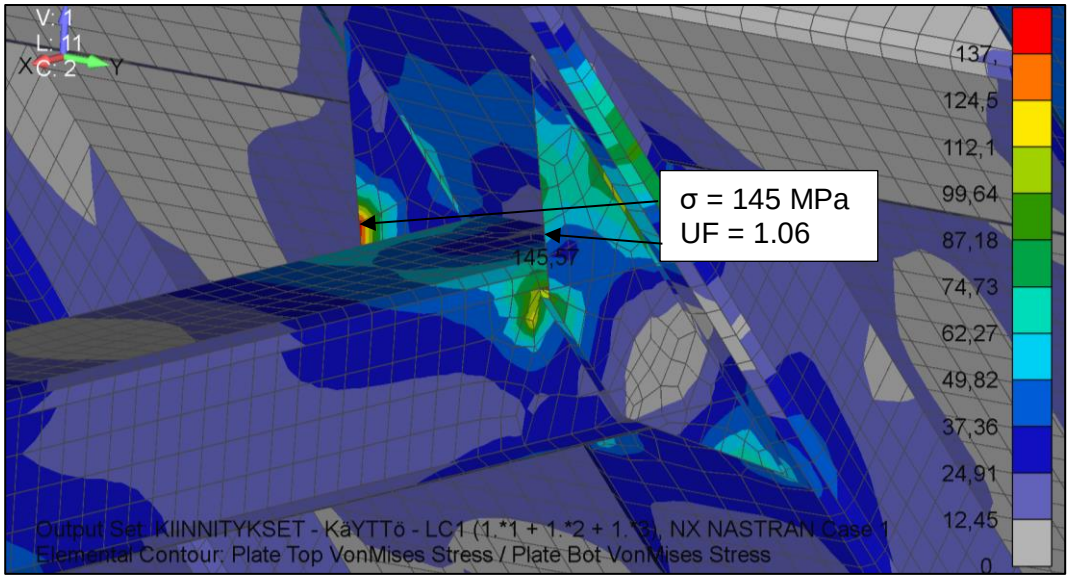


Figure 5.2.6. Von Mises stresses of the structure [MPa] in the operational load combination LC1, at the end of transverse beam.

Von Mises stress at the bottom front corner of the shell (near nozzle) is presented in figure 5.2.7. The maximum stress, 116 MPa, is below the design stress.

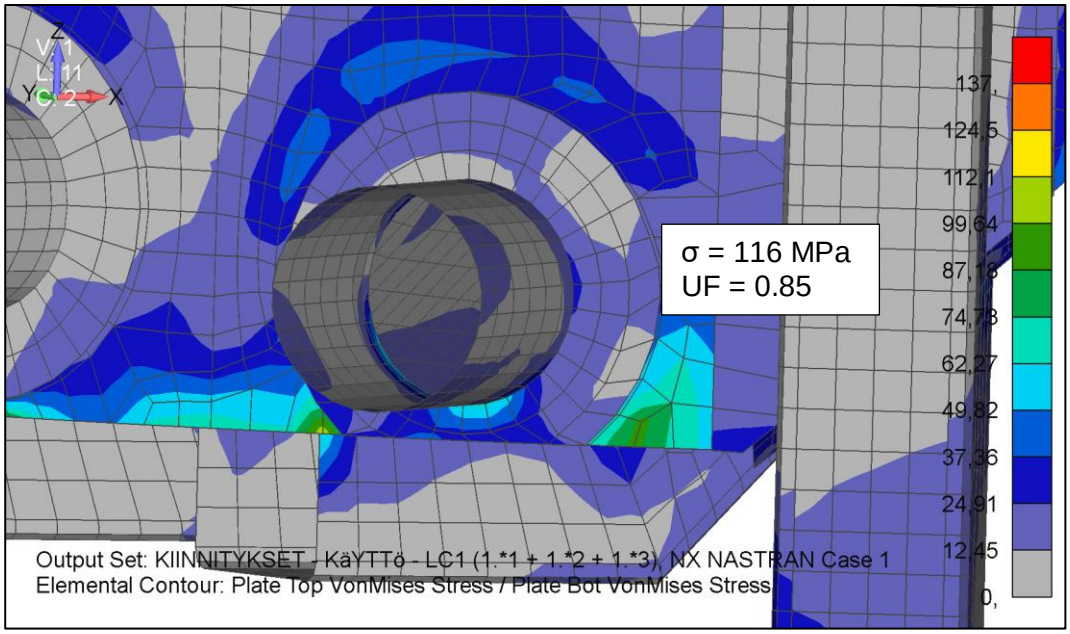


Figure 5.2.7. Von Mises stresses of the structure [MPa] in the operational load combination LC1, at the bottom front corner (near the nozzle) of the shell.

Stresses at the joint of the transverse and vertical RHS beam are presented in figure 5.2.8. The maximum stress, 124 MPa, is below the design stress.

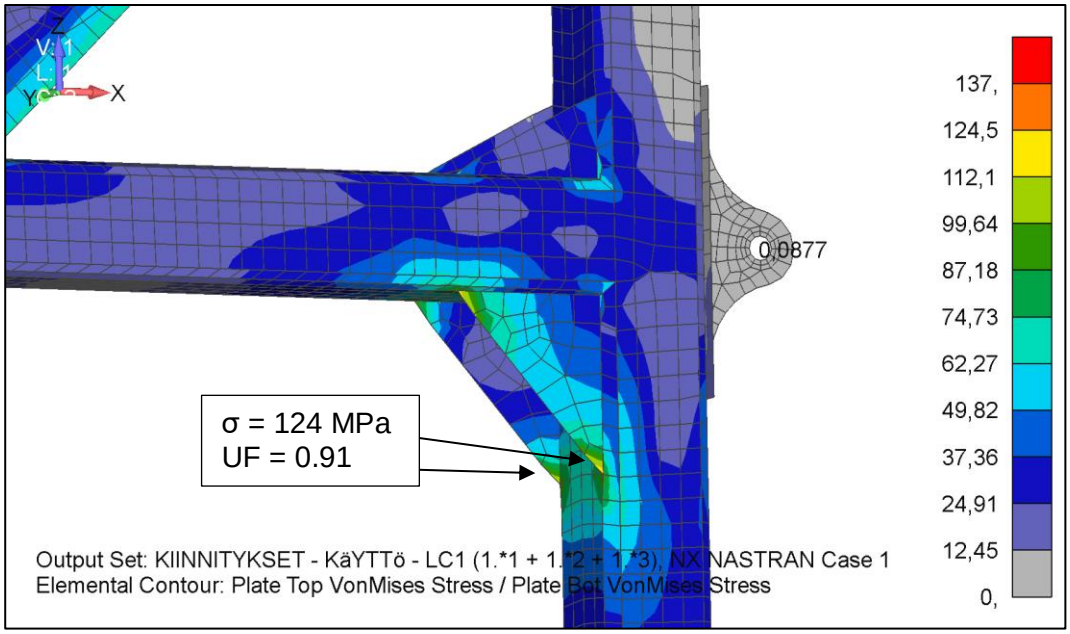


Figure 5.2.8. Von Mises stresses of the structure [MPa] in the operational load combination LC1, at the joint of the transverse and vertical RHS beam.

Von Mises stresses at the bottom part of the most critical leg are presented in figure 5.2.9. The stress peak 300 MPa locates at the edge of the bolt hole and it is unreal (rigid RBE2 element).

The local stress peak 203 MPa is unreal and the stress at the RHS-beam is calculated analytically based on axial force and bending of the leg. The calculated stress is 62 MPa and UF = 0.45.

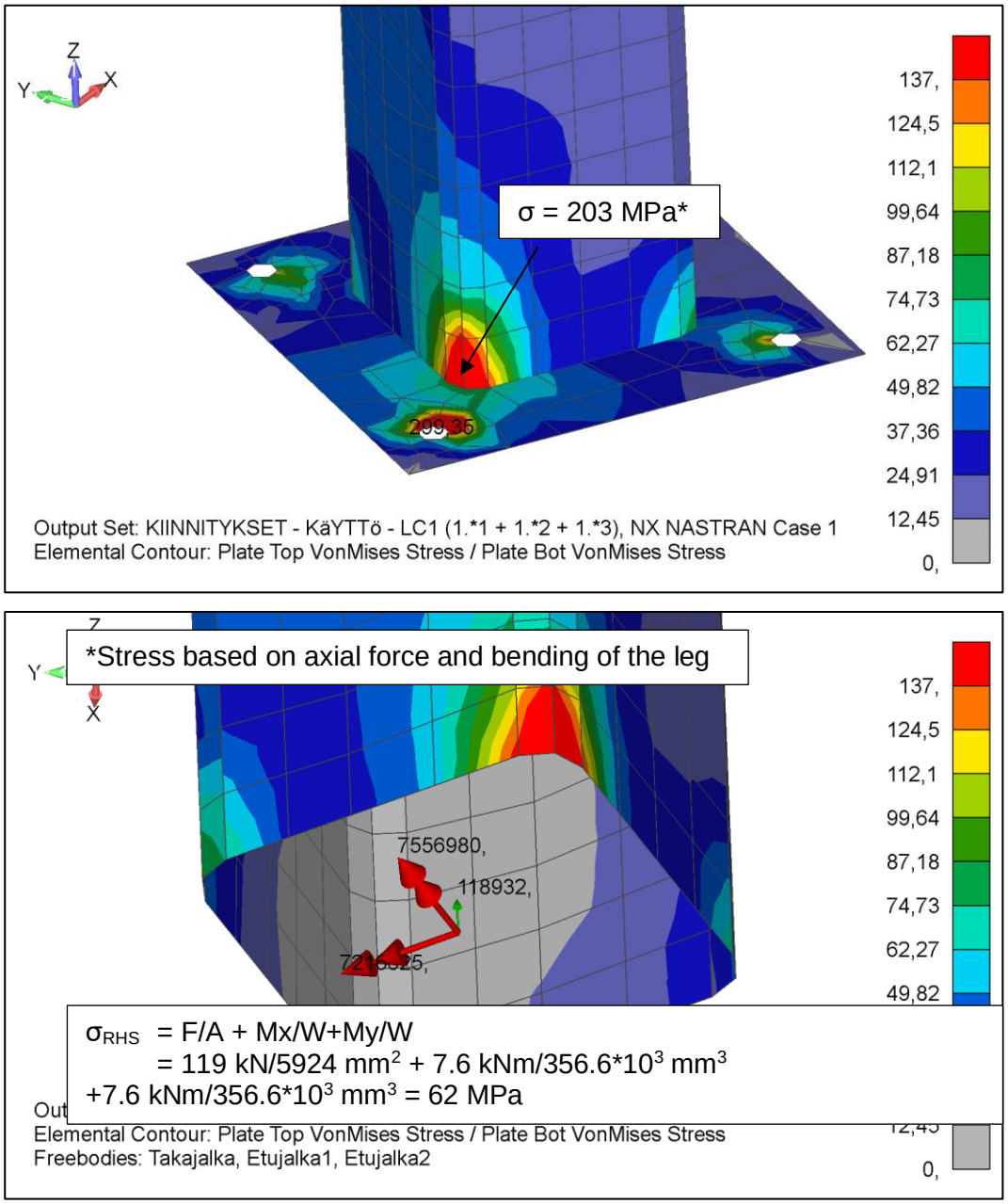


Figure 5.2.9. Von Mises stress [MPa] at the most critical leg.

The von Mises stresses of the structure at the static load cases LC2 (two different directions for the wind load) were well below the design stress 137 MPa maximum von Mises stress being 62 MPa (see fig 5.2.10).

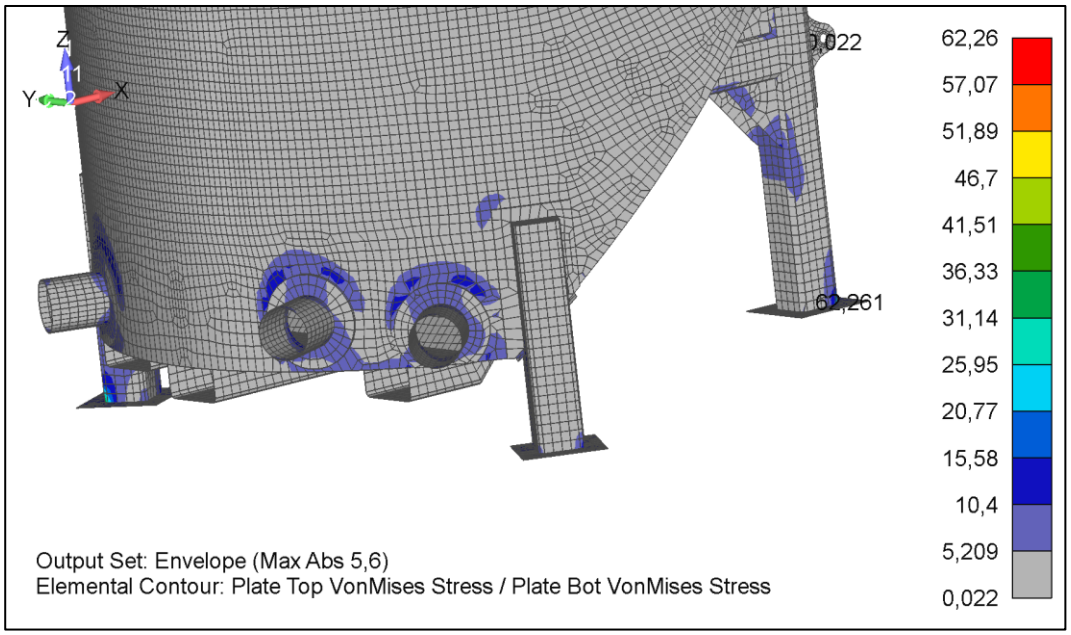


Figure 5.2.10. Maximum von Mises stress [MPa] at the two different load combinations of LC2.

5.3. Buckling analysis

Buckling analysis was performed for the load cases LC5 (two different wind load directions). The lowest load factor for the structure was 84 (>4), see figure 5.3.1.

The buckling strength is sufficient.

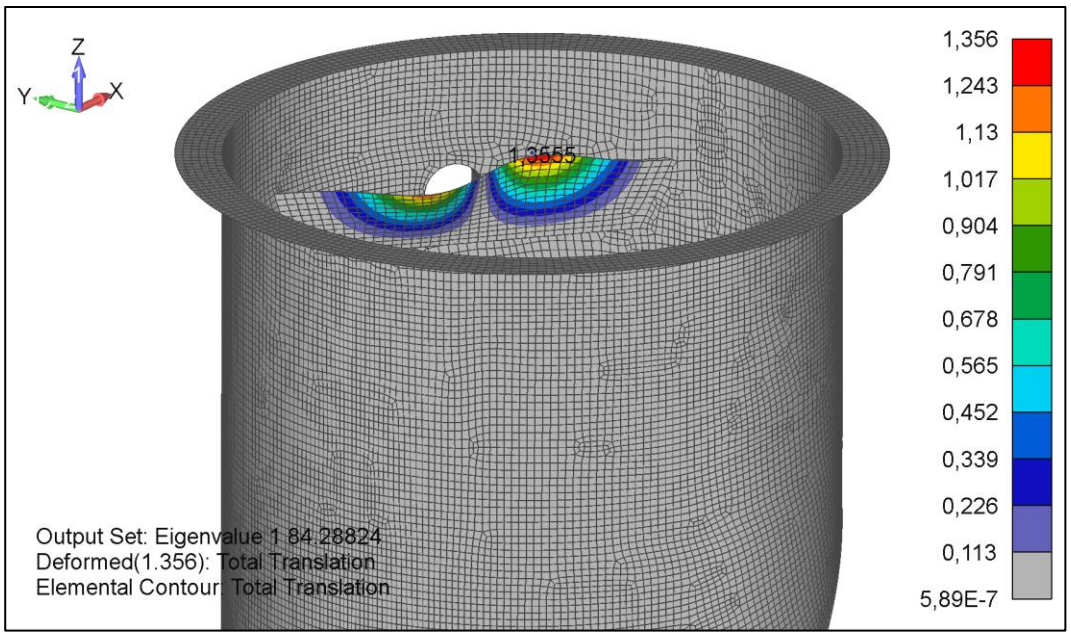


Figure 5.3.1. The lowest load factor in buckling analysis, with load case LC5.

5.4. Lifting lugs

Displacements in lifting situation are presented in figure 5.4.1. Maximum total translation is 3.5 mm in horizontal position lifting and 10.5 mm in vertical position lifting.

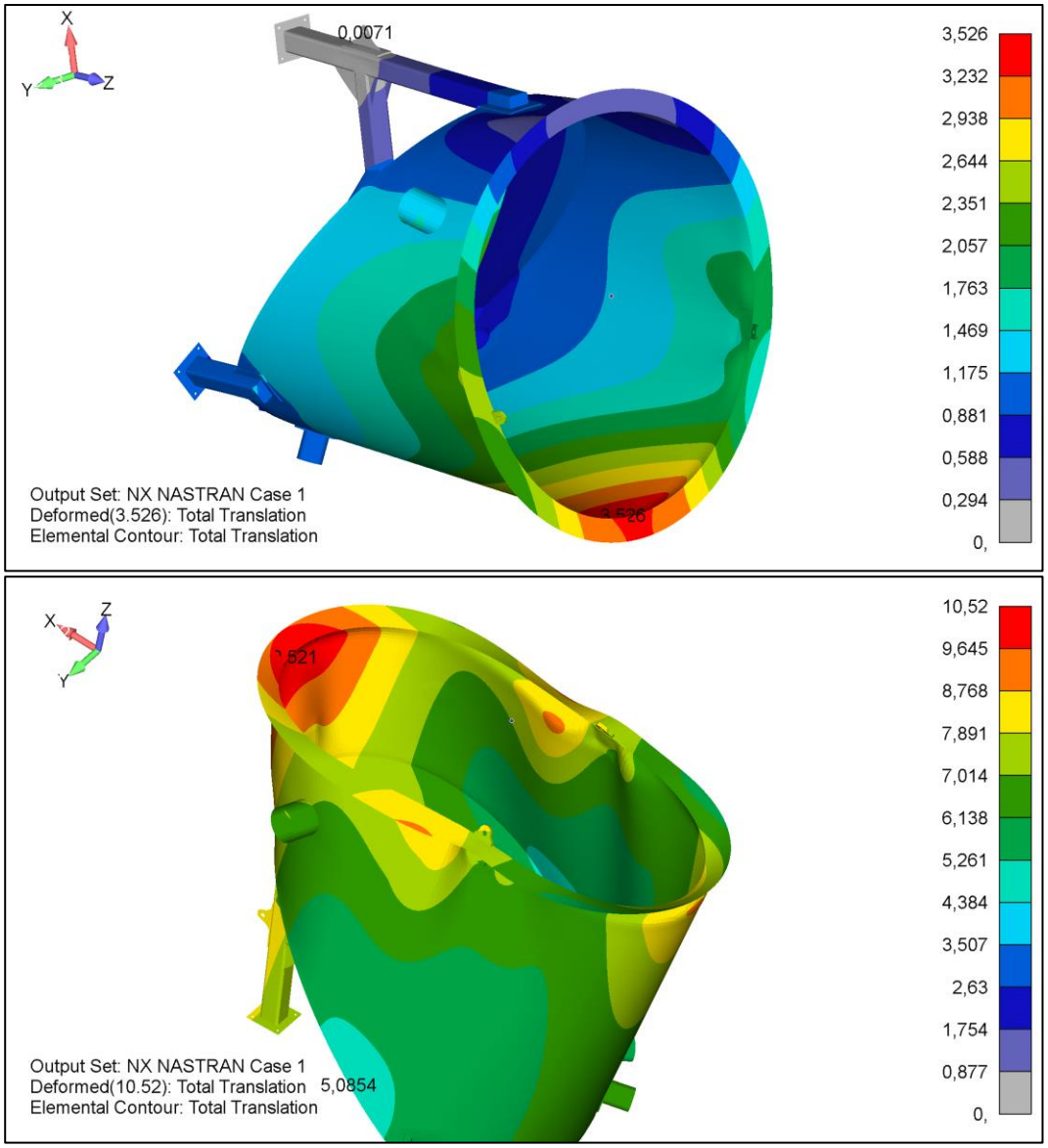


Figure 5.4.1. Total translation [mm] in lifting situations (Deformation is scaled by the factor 100.).

Von Mises stresses in the horizontal lifting are presented in figure 5.4.2. Maximum von Mises stress is 109 MPa at the lifting lug.

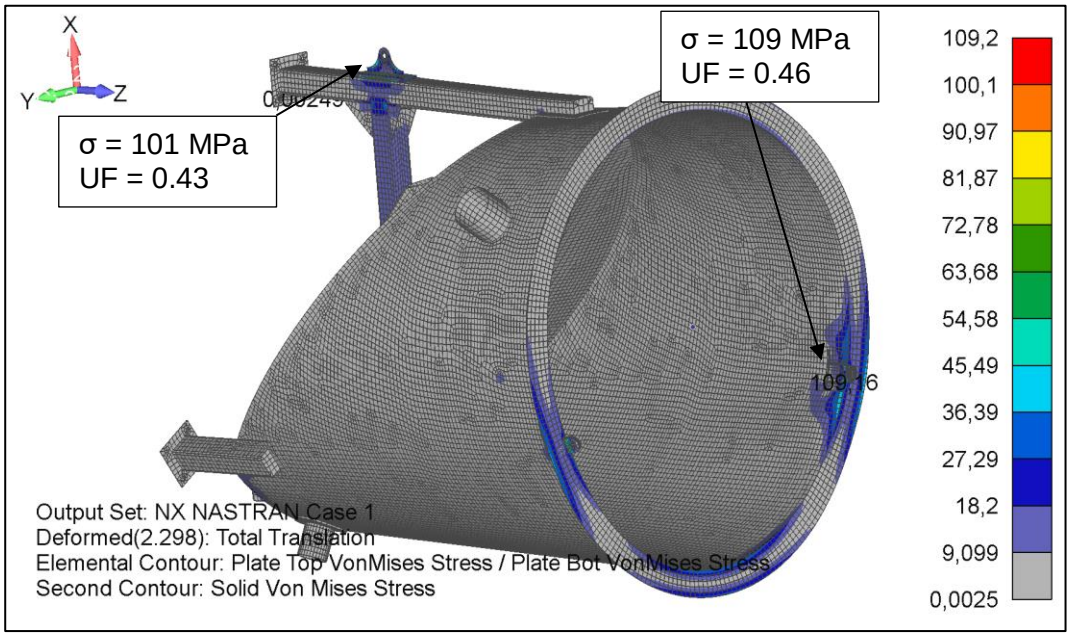


Figure 5.4.2. Von Mises stress [MPa] in lifting situation (tank in horizontal position)

Strength of the fillet weld (a5) at the attachment of the lug and reinforcement plate is calculated analytically. The length of the lifting lug weld is l = 2*380mm+2*25mm = 810 mm. The tension force for the lifting lug is F = 85 kN (see figure 5.4.3a). Therefore, the tension force at the fillet weld is F/(l*a) = 85000N / (810mm*5mm) = 21 MPa. For the weld at the reinforcement plate (between lug and the leg) a5 fillet weld is also strong enough (l_{reif} = 2*506mm+2*140mm = 1292 mm).

The analytical calculations for lifting lug strength with the lifting hook are presented in figures 5.4.3a and 5.4.3b. The yield strength of the lifting material being 235 MPa for shear the allowable stress is 235 MPa/√3 = 136 MPa. The total force of 85 kN is subjected to two shear planes and therefore σ_{shear} =F/2A = 85 kN/2000 mm^2 = 43 MPa (< 136 MPa).

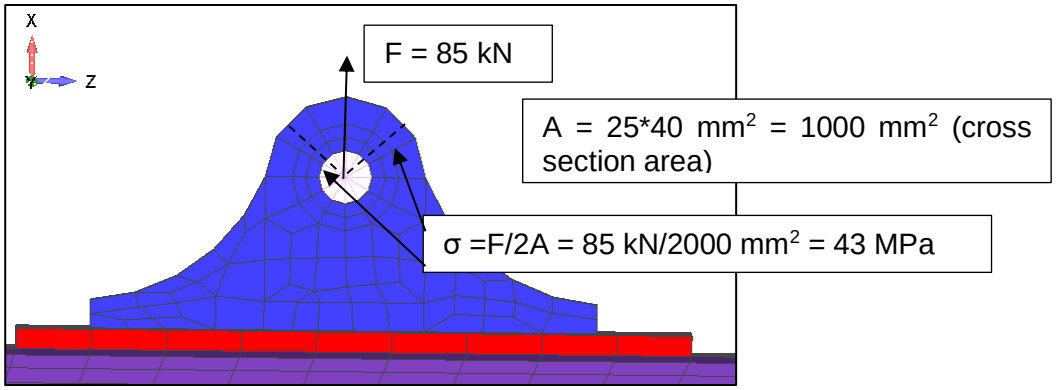


Figure 5.4.3a. Stress calculation for the cross section of the lifting lug

It is assumed that the contact of the lifting lug and the lifting hook is 90 deg. The area of the contact is therefore A_c = 25mm*π*40mm/4 = 1571 mm^2 and compressing stress at the surface of the lifting lug is 54 MPa (< 235 MPa).

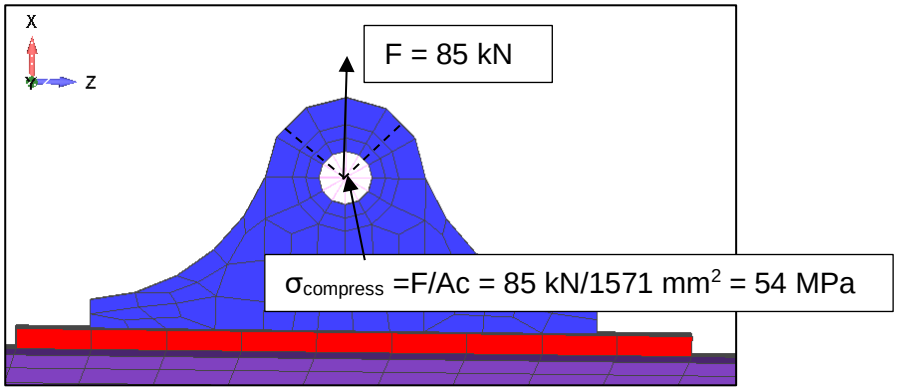


Figure 5.4.3b. Stress calculation for the compression between the lifting hook and the lifting lug.

Von Mises stresses in the vertical lifting are presented in figure 5.4.4. Maximum von Mises stress is 227 MPa at the tank shell upper edge.

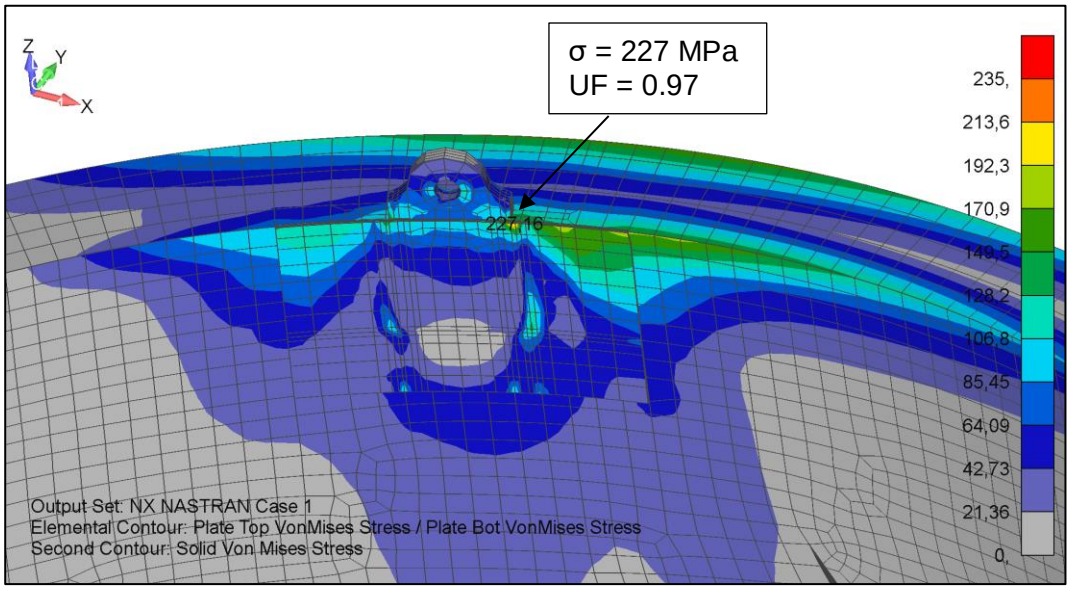


Figure 5.4.4. Von Mises stress [MPa] in lifting situation (tank in vertical position).

Strength of the fillet weld (a5) at the attachment of the upper lifting lugs is calculated analytically. The length of the weld (vertical parts) is l = 2*200 mm = 400 mm (see figure 5.4.5). The total force resultant for the lifting lug is F = 78 kN. Therefore, the shear stress at the fillet weld is F/(l*a) = cos (40°)*78000N / (400mm*5mm) = 30 MPa.

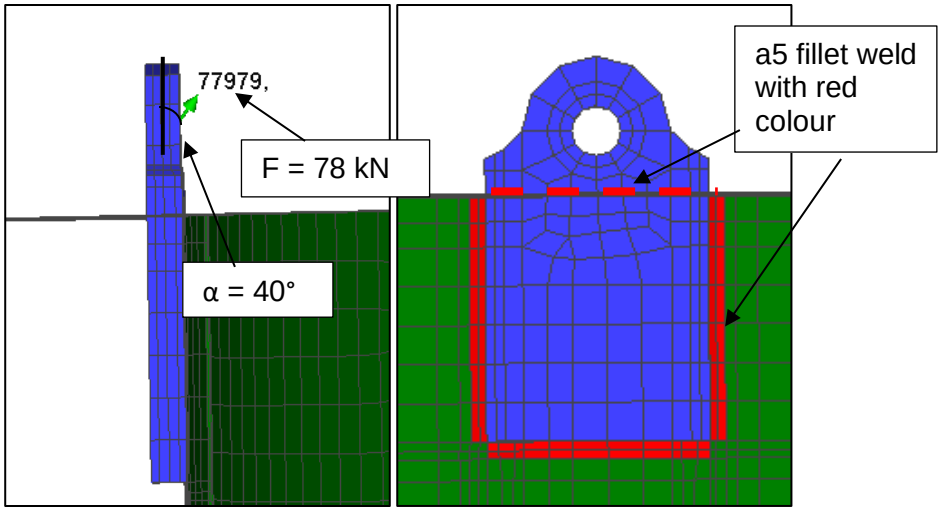


Figure 5.4.5. Total force for the fillet weld of the upper lifting lug.

5.5. Footing loads

Footing loads are presented in the global rectangular coordinate system (figure 5.5.1). The numbering of the legs is presented also in figure 5.5.1.

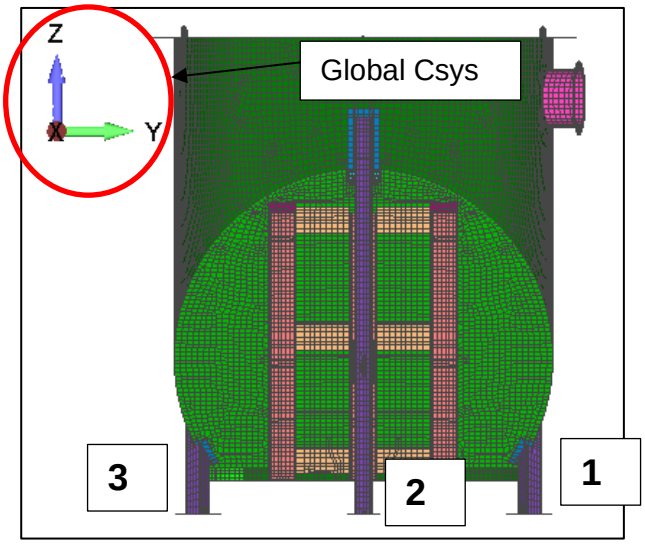


Figure 5.5.1. Orientation of the global rectangular coordinate system and numbering the feet.

Constraint forces [kN] and moments [kNm] in feet in seismic load combinations are presented in attachment 1.

Constraint forces and moments in feet in operational load cases LC1 and LC2 are presented in attachment 1.

The maximum constraint loads for footing in operation, seismic, wind and full of solids conditions are presented in Table 5.5.1.

Collected maximum values for footing loads (constraint) from operation, seismic and wind loads situations are presented in Table 5.5.2.

Table 5.5.1. Maximum constraint forces [kN] and moments [kNm] in feet in different conditions.

	Operation			Seismic		
Foot	1	2	3	1	2	3
FX	24.7	-49.1	24.5	28.1	50.4	27.9
FY	-18.8	0.1	18.7	18.5	0.4	21.6
FZ	117.9	74.1	119.1	119.0	82.5	122.0
MX	2.5	-0.1	-2.4	2.5	1.4	3.6
MY	9.5	-14.8	9.7	10.6	15.4	10.6
MZ	2.8	0.0	-2.7	3.2	0.0	3.2
	Wind			Full of solids		
Foot	1	2	3	1	2	3
FX	2.6	4.3	2.2	46.0	-91.3	45.6
FY	6.4	0.3	1.9	-33.2	0.1	33.1
FZ	26.2	20.1	17.0	206.3	131.8	205.6
MX	1.5	0.3	1.3	4.6	-0.1	-4.6
MY	0.9	1.5	0.8	17.7	-27.5	17.7
MZ	0.3	0.1	0.3	5.4	0.0	-5.2

Table 5.5.2. Maximum values for footing loads. (constraint forces [kN] and moments [kNm] in feet). (without full of solids loads).

	Maximum values		
Foot	1	2	3
Max shear FXY	33.6	50.4	33.4
Max axial FZ	119.0	82.5	122.0
Min axial FZ	15.6	13.2	13.3
Max MX	2.5	1.4	3.6
Max MY	10.6	15.4	10.6
Max MZ	3.2	0.1	3.2

6. CONCLUSIONS

- The static and buckling strength of the tank structure was assessed. The loads for the structure were self-weight, weight of the contents, additional nozzle load, wind load and seismic load. The tank was evaluated according to code API 650. The wind and seismic loads were based on code SBC 301 – Structural - loading and forces.
- The geometry of the structure was based on geometry file *3310-TNK-007 CLEANER TAILS PUMP SUMP 15m3.x_t,* but several modifications were made for the structure. The modifications are presented in section 3.1.
- In general, while interpreting the FEA results, following aspects were taken into account:
    - Local stress concentrations are ignored if the neighbouring areas have sufficient capacity to carry the external load without significant plastic deformations.
    - The allowable product design stress for the bending type of stress is 1.5 times the allowable product design stress for the membrane type of stress.
    - Secondary stresses (e.g. due to enforced motion) are ignored, and primary stresses are used for judging the eligibility of the structure.
    - In order to avoid the risk of cyclic plasticity, exceeding three times the product design stress is not allowed.
- The design stress 137 MPa was exceeded in few locations with the operational load case (LC1). The maximum von Mises stress was 207 MPa (UF 1.5) at the reinforcement plate of the U-beam below the bottom plate. Anyhow, as presented in the chapter 5.2, all the studied details fulfilled the design requirements, when the stress results were evaluated more closely.
- Buckling strength was found to be sufficient.
- Static strength for lifting load cases (2.5 times the weight of the structure) was found sufficient with the presented lifting lug structures.
- Footing loads were reported with operational, wind and seismic load combinations separately for the legs (see section 5.5 and Attachment 1).

REFERENCES

[1] API, Standard 650, Welded tanks for Oil Storage

[2] SBC, 301 Structural - Loading and Forces

[3] Structural Design Criteria – Process Buildings and Structures: *MD-101-SMEM-EG-ST-CRT-0001*

[4] Civil, Structural and Architectural Design Criteria: *MD-216-1000-EG-ST-CRT-1001 (900707_GAU01_ZZ01_CEC03_00001_OU100656130)*

[5] GEOTECHNICAL INVESTIGATION REPORT: *MD-216-49RA-EG-CV-RPT-1001 (900707_GAU01_AN03_CQS03_49001_OU100664631)*

[6] OUTLINE DRAWING - 3310-TNK-007: CLEANER TAILS PUMP SUMP (15M3) : MD-216-3310-EG-ME-GAD-1010 (*900707_GAU07_ZF01_MTA01_33106_OU100660892*)

ATTACHMENT 1 FOOTING LOADS

Footing loads in four different seismic load combinations are presented in Table A1.1.

Table A1.1. Constraint forces [kN] and moments [kNm] in feet in seismic load combinations.

	EQ1			EQ2			EQ3			EQ4		
Foot	1	2	3	1	2	3	1	2	3	1	2	3
FX	21.2	-50.4	21.0	21.5	-49.7	24.1	25.0	-48.3	27.5	28.1	-47.5	27.9
FY	-17.4	0.1	17.4	-14.3	0.4	21.1	-14.8	0.4	21.6	-18.5	0.1	18.4
FZ	112.3	82.5	111.8	108.9	79.2	118.6	112.3	72.4	122.0	119.0	69.1	118.5
MX	2.4	0.0	-2.4	1.3	-0.4	-3.5	1.4	-0.4	-3.6	2.5	0.0	-2.5
MY	8.2	-15.4	8.2	8.2	-15.1	9.4	9.4	-14.4	10.6	10.6	-14.1	10.6
MZ	2.5	0.0	-2.4	2.5	0.0	-2.8	2.8	0.0	-3.2	3.2	0.0	-3.1

Maximum loads in seismic analysis is presented in Table A1.2

Table A1.2. Maximum (absolute values) constraint forces [kN] and moments [kNm] in feet in seismic load combinations.

	Seismic load combinations		
Foot	1	2	3
FX	28.1	50.4	27.9
FY	18.5	0.4	21.6
FZ	119.0	82.5	122.0
MX	2.5	1.4	3.6
MY	10.6	15.4	10.6
MZ	3.2	0.0	3.2

Footing loads in operational load combinations are presented in Table A1.3 and maximum loadings with wind in Table A1.4.

Table A1.3. Constraint forces [kN] and moments [kNm] in feet in static load combinations.

	LC1 (operation)			LC2_1 (wind)			LC2_2(wind)		
Foot	1	2	3	1	2	3	1	2	3
FX	24.7	-49.1	24.5	-2.1	-4.3	-2.2	2.6	-2.7	0.1
FY	-18.8	0.1	18.7	-1.3	0.0	1.3	-6.4	-0.3	-1.9
FZ	117.9	74.1	119.1	15.6	20.1	17.0	26.2	13.2	13.3
MX	2.5	-0.1	-2.4	0.1	0.0	0.1	1.5	0.3	1.3
MY	9.5	-14.8	9.7	-0.9	-1.5	-0.8	1.2	-0.8	0.2
MZ	2.8	0.0	-2.7	-0.3	0.0	0.3	0.3	0.1	0.2

Table A1.4. Maximum constraint forces [kN] and moments [kNm] in feet in static load combinations with wind.

	Wind load combinations		
Foot	1	2	3
FX	2.6	4.3	2.2
FY	6.4	0.3	1.9
FZ	26.2	20.1	17.0
MX	1.5	0.3	1.3
MY	0.9	1.5	0.8
MZ	0.3	0.1	0.3

ATTACHMENT 3 STAAD INPUT FOR F5 DESIGN

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STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-May-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
494 -6.66 -1.67 -4.145; 495 -6.66 -2.87 -4.145; 496 -6.66 -0.82 -4.145;
497 -3.96 -1.67 -4.145; 498 -3.96 -2.87 -4.145; 499 -3.96 -0.82 -4.145;
519 -7.16 -2.87 -4.895; 520 -7.16 -2.87 -4.845; 521 -7.11 -2.87 -4.895;
522 -7.11 -2.87 -4.845; 523 -7.06 -2.87 -4.895; 524 -7.06 -2.87 -4.845;
525 -7.01 -2.87 -4.895; 526 -7.01 -2.87 -4.845; 527 -6.96 -2.87 -4.895;
528 -6.96 -2.87 -4.845; 529 -6.91 -2.87 -4.895; 530 -6.91 -2.87 -4.845;
531 -6.86 -2.87 -4.895; 532 -6.86 -2.87 -4.845; 533 -6.81 -2.87 -4.895;
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537 -6.71 -2.87 -4.895; 538 -6.71 -2.87 -4.845; 539 -6.66 -2.87 -4.895;
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543 -6.56 -2.87 -4.895; 544 -6.56 -2.87 -4.845; 545 -6.51 -2.87 -4.895;
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552 -6.36 -2.87 -4.845; 553 -6.31 -2.87 -4.895; 554 -6.31 -2.87 -4.845;
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573 -5.81 -2.87 -4.895; 574 -5.81 -2.87 -4.845; 575 -5.76 -2.87 -4.895;
576 -5.76 -2.87 -4.845; 577 -5.71 -2.87 -4.895; 578 -5.71 -2.87 -4.845;
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*START GROUP DEFINITION
*FLOOR
*END GROUP DEFINITION
UNIT METER KN
ELEMENT PROPERTY
564 TO 637 654 TO 727 744 TO 817 834 TO 907 924 TO 997 1014 TO 1087 -
1104 TO 1177 1194 TO 1267 1284 TO 1357 1374 TO 1447 1464 TO 1537 -
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2004 TO 2077 2094 TO 2167 2184 TO 2257 2274 TO 2347 2364 TO 2437 -
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2904 TO 2977 2994 TO 3067 3084 TO 3157 3174 TO 3247 THICKNESS 0.6
UNIT METER MTON
SUPPORTS
564 TO 637 654 TO 727 744 TO 817 834 TO 907 924 TO 997 1014 TO 1087 -
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3247 PLATE MAT DIRECT Y SUBGRADE 4129.85 PRINT
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007

POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.21467e+006
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DENSITY 2.40261
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DAMP 0.05
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G 946438
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END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
551 TO 554 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
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LOAD 1 LOADTYPE Dead TITLE DL_OPRATING
UNIT METER KN
SELFWEIGHT Y -1
JOINT LOAD
496 FX 18.7 FY -119.1 FZ 24.5 MX 9.7 MY -2.7 MZ -2.4
499 FX -18.8 FY -117.9 FZ 24.7 MX 9.5 MY 2.8 MZ 2.5
LOAD 2 LOADTYPE Wind TITLE WL
JOINT LOAD
496 FX 1.9 FY -17 FZ 2.2 MX 0.8 MY 0.3 MZ 1.3
499 FX 6.4 FY -26.2 FZ 2.6 MX 0.9 MY 0.3 MZ 1.5
LOAD 3 LOADTYPE Seismic TITLE EL
JOINT LOAD
496 FX 21.6 FY -122 FZ 27.9 MX 10.6 MY 3.2 MZ 3.6
499 FX 18.5 FY -119 FZ 28.1 MX 10.6 MY 3.2 MZ 2.5
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LOAD COMB 101 1.4 DL
1 1.4
LOAD COMB 102 1.2 DL + 1.6 WL
1 1.2 2 1.6
LOAD COMB 103 1.2 DL + 1.0 EL
1 1.2 3 1.0
LOAD COMB 104 0.9 DL + 1.6 WL
1 0.9 2 1.6
LOAD COMB 105 0.9 DL + 1.0 EL
1 0.9 3 1.0
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****Serviceanbility check

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LOAD COMB 201 1. DL
1 1.0
LOAD COMB 202 1 DL + 1 WL
1 1.0 2 1.0
LOAD COMB 203 1 DL + 0.7 EL
1 1.0 3 0.7
LOAD COMB 204 0.6 DL + 1 WL
1 0.6 2 1.0
LOAD COMB 205 0.6 DL + 0.7 EL
1 0.6 3 0.7
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PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
UNIT METER KN
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
FC 28000 ALL
FYMAIN 414000 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN COLUMN ALL
END CONCRETE DESIGN
*
FINISH
```